CORE MATHEMATICS GRADE 10: STATISTICS I

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 3.0: Statistics and Probability
Sub-Strand	Sub-Strand 3.1: Statistics I
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Collection of Data — meaning of data, sources of data, and appropriate methods of data collection and recording – Frequency Distribution Table — constructing frequency distribution tables for ungrouped data – Frequency Distribution Table — grouping data into classes and constructing frequency distribution tables for grouped data – Mean, Mode and Median of Ungrouped Data — calculating measures of central tendency for ungrouped data sets – Mean, Mode and Median of Grouped Data — calculating measures of central tendency (including modal class) for grouped data – Representation of Data — drawing histograms with equal class widths from grouped data – Representation of Data — drawing histograms with unequal class widths and frequency polygons from grouped data – Interpretation of Data — reading, analysing and making conclusions from histograms and frequency polygons to support informed decision making
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) collect data from real-life sources, b) draw a frequency distribution table for grouped and ungrouped data, c) determine mean, mode and median of grouped and ungrouped data, d) represent data using histograms and frequency polygons, e) interpret data from histograms and frequency polygons, f) promote data collection, organisation and interpretation to support informed decision making.
Core Competencies	– Critical Thinking and Problem Solving: the learner uses histograms and frequency polygons to examine data and make conclusions about real-life situations in the Kenyan community. – Self-Efficacy: the learner discusses with peers the meaning of the term data and explores local data sources such as market prices at Wakulima Market, rainfall records at Kenya Meteorological Department stations, or maize yield data from Rift Valley farmers. – Digital Literacy: the learner uses digital devices to draw histograms and frequency polygons and to make data-driven conclusions. – Communication and Collaboration: the learner presents and shares findings from data to different groups of people such as school administrators, parents or local community leaders.

Core Values	– Integrity: the learner collects and records data accurately, ensuring that figures such as daily maize prices or student scores are not altered or fabricated. – Unity: the learner works in groups and uses collected data to draw histograms and frequency polygons, respecting each member's contribution. – Love: the learner is considerate of others as they discuss in pairs and identify the modal class for grouped data, listening actively and affirming peers' ideas.
Science & Engineering Practices	– Analysing and Interpreting Data: the learner organises raw data into frequency distribution tables, computes measures of central tendency, and draws histograms and frequency polygons to identify patterns and trends in Kenyan community data sets. – Asking Questions and Defining Problems: the learner formulates statistical questions about real-life situations — such as "What is the most common daily expenditure on food in a Nairobi household?" — to motivate data collection and analysis. – Constructing Explanations: the learner uses computed statistics (mean, mode, median) and graphical representations (histograms, frequency polygons) to explain observed patterns in data and communicate findings to an audience. – Obtaining, Evaluating and Communicating Information: the learner identifies appropriate data sources, evaluates the quality and relevance of collected data, and communicates conclusions drawn from graphical representations to peers and community stakeholders.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Decision Making): the learner makes informed decisions from data interpretation — for example, a school canteen decides which food items (ugali, githeri, mandazi) to stock most based on learner preference data displayed in a frequency polygon. – Parental Engagement and Empowerment: the learner seeks support and guidance from parents or guardians to collect, organise and present data on an identified situation in the school or community, such as household water usage or daily food expenditure. – Road Safety (Community Awareness): the learner collects and analyses data on the availability of pedestrian crossing points along major roads in the locality, then uses histograms to present findings that advocate for safer road infrastructure.
Career Connections	– Data Analyst / Statistician: this sub-strand builds foundational skills in data collection, organisation and graphical representation used daily by statisticians at institutions such as the Kenya National Bureau of Statistics (KNBS). – Public Health Officer: epidemiologists and public health officers in Kenya use frequency distributions and histograms to track disease incidence — e.g., malaria cases per county — and guide resource allocation. – Agricultural Economist / Agronomist: professionals working with organisations like the Kenya Agricultural and Livestock Research Organisation (KALRO) use statistical tables and graphs to analyse crop yields (maize, tea in Kericho, sugarcane in Mumias) and advise farmers. – Journalist / Media Analyst: Kenyan journalists and infographic designers use histograms and frequency polygons to present election results, census data and economic indicators to the public. – Engineer / Urban Planner: engineers and planners at bodies like the Kenya Urban Roads Authority (KURA) use traffic data histograms to design roads and pedestrian infrastructure in Nairobi and Mombasa.
Focus for Lessons	This unit introduces Grade 10 learners to the foundational concepts of Statistics through the real-life lens of data that surrounds them in Kenya — from market prices and rainfall records to school performance and road safety. Learners progress systematically from collecting and organising raw data into frequency distribution tables, through computing measures of central tendency for both ungrouped and grouped data, to constructing and interpreting histograms and frequency polygons. The pedagogical approach is collaborative and inquiry-driven: learners generate their own data sets from the school or local community, wrestle with how best to represent and summarise those data, and ultimately use their statistical findings to make and justify real decisions — mirroring the work of statisticians, public health officers and agricultural economists across Kenya.

Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we use statistics to tell an honest and useful story about data from our own Kenyan community — and help people make better decisions? KICD KEY INQUIRY QUESTIONS: 1. What is statistics? 2. How do we represent data? 3. How do we use statistics in day-to-day life?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Maize Price Mystery at Wakulima Market" Every week, a Nairobi family sends someone to Wakulima Market to buy maize flour. Over the past month the family has noticed that the price they pay per 2 kg packet seems to vary a lot — some days it feels cheap, other days shockingly expensive — yet their neighbour who shops at the same market says the price is "always about the same." Both cannot be right, yet both are telling the truth as they experienced it. The teacher displays a handwritten list of 40 daily maize flour prices (in Kenyan shillings) collected over eight weeks from a stall at Wakulima Market. The raw list looks chaotic — numbers jump up and down with no obvious pattern. Students are asked: Who is right — the family that says prices vary wildly, or the neighbour who says prices are stable? How would you use these 40 numbers to settle the argument fairly? This is genuinely puzzling because: (i) a single glance at the raw data gives no clear answer — the need for organisation is immediately felt; (ii) different summary statistics (mean vs. median vs. mode) can give contradictory impressions, creating productive cognitive conflict; (iii) a histogram of the same data suddenly makes the distribution visible — a powerful "aha" moment that motivates graphical representation; (iv) the context is universally relatable to Kenyan learners who participate in household food purchasing, making the mathematics feel urgent and meaningful rather than abstract.
Storyline Thread	Lesson 1 — "What is this data trying to tell us?" (Anchoring Phenomenon Launch) Learners encounter the Wakulima Market maize price list for the first time. In pairs they predict: Is the price stable or variable? They attempt to answer using only the raw list — discovering how difficult it is. The class constructs a shared Driving Question Board (DQB). Key vocabulary introduced: data, raw data, data sources, tally, frequency. Learners begin collecting their own class data set (e.g., time taken to walk to school, or number of meals eaten per day) to use throughout the unit. Lesson 2 — "Organising the chaos: Frequency Distribution Tables for Ungrouped Data" Learners use the maize price data (and their own class data set) to construct frequency distribution tables for ungrouped data. They discover that organisation immediately reveals patterns invisible in the raw list. The concept of frequency, tally charts and the frequency column are formalised. Learners revisit their Predict from Lesson 1 — does the table help settle the family vs. neighbour argument yet? Lesson 3 — "Grouping data into classes: Frequency Distribution Tables for Grouped Data" The maize price data is now grouped into class intervals (e.g., Ksh 40–49, 50–59, etc.). Learners debate: How wide should a class

	be? What do we lose by grouping? The modal class is identified and discussed. Learners construct grouped frequency tables for the Kisumu rainfall supporting phenomenon and compare the two representations (ungrouped vs. grouped). The DQB is updated with new questions that have emerged. Lesson 4 — "What is the 'middle' of our data? Mean, Mode and Median for Ungrouped Data" Using the ungrouped maize price data and the class-collected data set, learners calculate mean, mode and median by hand. The teacher uses the mandazi canteen supporting phenomenon to provoke discussion: which measure is most honest? Learners discover that the mean can be distorted by extreme values (e.g., a day when prices spiked due to a shortage in Rift Valley). The three measures are compared and their appropriate uses discussed. Lesson 5 — "Measures of central tendency for Grouped Data" Learners extend their understanding to compute the mean (using midpoints), identify the modal class, and estimate the median for grouped data from the frequency distribution tables built in Lesson 3. The Kisumu rainfall data is used as a second worked example. Learners discuss: Why can we only estimate the mean and median for grouped data? The DQB is updated — some earlier questions are now answered, new ones emerge about representation. Lesson 6 — "Drawing Histograms (Equal Class Widths)" Learners are introduced to histograms as the appropriate graphical tool for grouped continuous data. Using the grouped maize price frequency table, learners draw histograms on graph paper (frequency on the y-axis, class intervals on the x-axis) with equal class widths. The distinction between a histogram and a bar chart is emphasised. Learners also draw the histogram for the pedestrian crossing data (supporting phenomenon 4) and describe its shape (skewed). Lesson 7 — "Histograms with Unequal Class Widths and Frequency Polygons" Learners encounter grouped data where class widths are not equal (e.g., matatu journey time data from supporting phenomenon 2 may have a combined "over 60 minutes" class). Frequency density (frequency ÷ class width) is introduced as the correct y-axis measure for unequal-width histograms. Learners then overlay a frequency polygon on their histogram by joining the midpoints of each bar. The shape of the distribution (symmetric, skewed, bimodal) is read and interpreted. DQB is updated. Lesson 8 — "Reading the story in the graph: Interpreting Histograms and Frequency Polygons + Unit Project Presentations" Learners practise interpreting histograms and frequency polygons to extract information: estimating frequencies for given intervals, identifying the modal class, comparing two distributions drawn on the same axes. The lesson culminates in brief group presentations of the unit project — each group has collected real data from the school or community (canteen sales, school arrival times, water usage), organised it, computed statistics, drawn a histogram or frequency polygon, and used their findings to make a recommendation to a real audience (e.g., the school canteen manager or a local road-safety committee). The DQB is reviewed: Which questions have been fully answered? Which remain open for future study (e.g., Strand 3.2 Probability)?
Supporting Phenomena	– SUPPORTING PHENOMENON 1 — "How long do Nairobi matatu journeys really take?" A learner records the journey time (in minutes) for 30 matatu trips from Westlands to CBD over two weeks. The times range from 12 minutes to 74 minutes. Why is there such a spread? Where does most of the data cluster? A frequency polygon of the journey times reveals a bimodal shape — peak hour vs. off-peak — introducing the idea that a single average can be misleading and that the shape of a distribution tells a richer story. – SUPPORTING PHENOMENON 2 — "Rainfall in Kisumu: is it getting wetter?" The Kenya Meteorological Department records show monthly rainfall totals (mm) for Kisumu over the past five years (60 data points). Farmers near Lake Victoria say recent rains are "heavier than before," but a government official says "average rainfall is unchanged." A grouped frequency table and

	histogram of the data lets learners investigate both claims — connecting statistics to agriculture and climate, and illustrating the difference between the mean and the spread of a distribution. – SUPPORTING PHENOMENON 3 — "Which class sells the most mandazi at the school canteen?" The canteen records daily mandazi sales per class over 20 school days. The data set has an obvious mode (the class that buys the most) but the mean is pulled upward by one or two very high-sales days. Learners discover that the mode is more useful here than the mean — motivating a nuanced understanding of when each measure of central tendency is appropriate. – SUPPORTING PHENOMENON 4 — "Pedestrian crossings along Thika Superhighway." A community group counts the number of designated pedestrian crossing points per 1 km stretch along 25 different segments of Thika Superhighway. Most segments have very few crossings; a small number of segments (near shopping malls) have many. A histogram of this data produces a positively skewed distribution, and learners interpret what it means for road safety — directly connecting mathematics to a contemporary Kenyan public issue.

LESSON 1 (80 minutes): What Is This Data Trying to Tell Us? — Launching the Maize Price Mystery	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon by immersing learners in a real, unresolved data puzzle from Wakulima Market, surfacing prior knowledge about data, and generating the driving questions that will guide the entire sub-strand.
Knowledge	Learners will know that: (i) data is a collection of facts or measurements gathered from real-life sources; (ii) raw, unorganised data is difficult to interpret; (iii) different ways of summarising or displaying data can lead to different — even contradictory — impressions of the same situation.
Skills	Learners will be able to: (i) read and describe a raw data set; (ii) make and justify a prediction about what data shows before formal analysis; (iii) pose statistical questions that could settle a real-life disagreement; (iv) begin to classify questions about the data onto a Driving Question Board.
Attitudes	Learners will appreciate that: (i) mathematics is a tool for resolving real community disputes fairly and honestly; (ii) jumping to conclusions from raw data is unreliable — organisation and representation are necessary; (iii) working collaboratively with peers to question data builds integrity and social responsibility.
Key Inquiry Question	What is statistics? | How do we represent data? | How do we use statistics in day-to-day life?
Purpose in Storyline	This is Lesson 1 — the phenomenon launch. Learners encounter the Maize Price Mystery at Wakulima Market for the first time: a family says maize flour prices vary wildly; their neighbour says prices are always the same. Both are telling the truth as they experienced it. Learners wrestle with 40 raw daily prices, discover that raw data gives no clear answer, and generate the questions that will drive all remaining lessons in the sub-strand. The DQB is opened for the first time.
Safety Notes	No physical hazards. Ensure classroom discussion remains respectful — some learners' families may be directly affected by food price fluctuations. Avoid singling out learners based on household economic circumstances.

B. LESSON OVERVIEW

Learners are introduced to the anchoring phenomenon — 'The Maize Price Mystery at Wakulima Market' — through a handwritten list of 40 daily maize flour prices collected over eight weeks. The teacher presents the real-life disagreement between a family (prices vary wildly) and their neighbour (prices are always the same) and challenges learners to decide who is right using only the raw numbers. Working in pairs and small groups, learners first predict an answer, then struggle productively with the raw list, discovering for themselves that unorganised data cannot settle the argument. The lesson closes by opening the class Driving Question Board (DQB), onto which learners post the questions they now want answered. No formal statistical procedures are taught in this lesson; the goal is to create an urgent, felt need for the tools that will be learned in subsequent lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 —	The teacher reveals the Wakulima Market scenario verbally and in	1. Read the scenario aloud dramatically, pausing after 'Both	Think-Pair-Share with public commitment: learners commit to a	Observable indicator: every learner has written a prediction

Predict 📹 **VIDEO:** Statistics: Alternate variance formulas Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	writing on the board: a Nairobi family notices maize flour prices jumping up and down over eight weeks; their neighbour insists prices are 'always about the same.' Both shop at the same stall. Learners are given 60 seconds of silent thinking time, then write a personal prediction in their exercise books answering: 'Who do you think is right — the family or the neighbour? Give one reason for your prediction.' Pairs then share predictions with each other (think-pair) before four learners are cold-called to share with the class. Predictions are written on the board under two columns: 'Team Family — Prices Vary' and 'Team Neighbour — Prices Stable' with tally marks showing class split. The teacher does NOT reveal the answer.	cannot be right — yet both are telling the truth as they experienced it.' Ask: 'How is it possible for two people who shop at the same market to have completely opposite experiences of the same prices?' [WAIT 12 seconds — do not fill the silence.] 2. Instruct: 'In your exercise book, write your prediction and ONE reason. You have 60 seconds — go.' Circulate and read 4–5 responses without commenting. 3. After pair-share, cold-call exactly 4 learners — deliberately pick two from each 'team.' Use names: 'Akinyi, tell us your prediction and why.' 'Omondi, do you agree or disagree with Akinyi?' 4. Record predictions on the board in two columns. Count and announce the class split: 'So 18 of us think prices vary, 14 think they're stable. How are we going to find out who is right?' [WAIT 10 seconds.] 5. Affirm all predictions: 'Every prediction here is reasonable — that is exactly why we need mathematics to settle this fairly.'	prediction before seeing data, activating prior intuitions about price variation, averages, and fairness. Making predictions public (board tally) creates a stake in the outcome and productive motivation to resolve the conflict using mathematics. This mirrors how scientists form hypotheses before collecting evidence.	with a reason in their exercise book before pair-share begins. Teacher scans books during circulation — look for learners who write 'I don't know' and prompt with: 'Think about the last time you went to the market — did you ever notice the price of unga was different from the week before?' Cold-call responses reveal whether learners have intuitive notions of average, range, or variability that can be built on.
Phase 2 — Observe 📹 **VIDEO:** Statistics: Alternate variance formulas Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook)	The teacher distributes a printed A4 sheet (or writes on the board) containing the 40 raw daily maize flour prices (in KES for a 2 kg packet) collected over 8 weeks from a Wakulima Market stall. The prices are presented in the order they were collected — deliberately chaotic and unorganised. Example data set: 92, 105, 88, 110, 95, 103, 89, 112, 98, 107, 91, 104, 86, 115, 100, 96, 108, 93, 111, 99, 87, 106, 94, 113, 101, 90, 109, 97, 114, 102, 88, 107, 95, 110, 91, 104, 98, 112, 86, 103. In groups of four, learners are given 10 minutes to study ONLY the raw list and attempt to answer: (A) What is the	1. Distribute or display the raw data list. Say: 'These are real prices — 40 days, 8 weeks, one stall at Wakulima Market. This is the evidence. Your job right now is NOT to calculate anything — just look at the list and try to answer the four questions on your sheet.' [WAIT 8 seconds for learners to read the instructions before starting the timer.] 2. Circulate during group work. Listen for productive struggle — groups should feel frustrated. If a group says 'this is impossible,' respond: 'What makes it feel impossible? Write that feeling down — it is an important observation.' Do NOT	Productive Struggle with Raw Data: learners experience firsthand the cognitive difficulty of extracting meaning from unorganised data. This is not a lesson failure — it is the designed insight. The frustration of not being able to definitively answer question C from raw data creates an authentic, felt need for frequency distribution tables, measures of central tendency, and graphical representation. The teacher names this explicitly at the end of the phase: 'The discomfort you just felt — that is exactly why statisticians developed tools to organise data. We are going to	Observable indicators: (i) All groups can correctly identify the minimum (86 KES) and maximum (115 KES) — if a group cannot, they need support with reading/scanning the list systematically; (ii) Groups disagree on question C — this disagreement is the desired outcome and signals readiness for the next phase; (iii) Listen for groups who spontaneously try to organise the data (sort, tally, count) — these learners are demonstrating prior statistical intuition; note them for cold-calling in later lessons. Red flag: groups who are completely silent and not

Source: CK-12 🔍 Search ARES for similar readings	cheapest price recorded? (B) What is the most expensive price recorded? (C) Without calculating anything — just by looking — would you say prices are mostly stable or mostly varying? (D) Could you use this raw list to convince someone definitively? Groups record their answers on a half-sheet of paper. After 10 minutes, groups share findings in a whole-class 'data gallery' — teacher reads out group answers and marks agreements/disagreements on the board.	hint at solutions. 3. After 10 minutes, facilitate sharing. Cold-call one reporter per group for questions A and B (cheapest/most expensive — these are findable). For question C, cold-call at least 3 groups and record disagreements: 'Group 3 says prices are varying. Group 5 says they look stable. Both groups looked at the SAME list — why might they see it differently?' [WAIT 12 seconds.] 4. For question D, press: 'Raise your hand if your group felt confident you could convince someone using only this raw list.' [Likely few hands.] 'Why not? What is missing?' Record student language on the board — words like 'organised,' 'pattern,' 'average,' 'chart' are gold — circle them.	learn every one of those tools.'	engaging — check for language barriers or low confidence and provide the prompt card: 'Start by circling the smallest number you can find.'
Phase 3 — Explain 📹 **VIDEO:** Statistics: Alternate variance formulas Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher now facilitates a whole-class sense-making discussion: 'What would need to happen to this list before we could use it to settle the argument fairly?' Learners brainstorm and the teacher records ideas. Then the teacher introduces — through guided questioning, NOT lecturing — three key concepts as vocabulary only (no procedures yet): (1) the idea of organising data into a frequency table; (2) the idea that a single summary number (like an average price) could represent all 40 prices; (3) the idea that a picture (graph) of the data might reveal patterns the raw list hides. The teacher demonstrates this with a powerful 'aha' moment: using the board, the teacher rapidly sorts the 40 prices into five rough bands (86–91, 92–97, 98–103, 104–109, 110–115) and draws a rough hand-sketch histogram. The class observes that	1. Ask: 'If a magistrate at a Nairobi court needed to use this data as evidence, what would they need us to do to it first?' [WAIT 12 seconds. Cold-call 3 learners.] Record their ideas — expect: 'sort it,' 'add it up,' 'find the middle,' 'draw a chart.' Affirm all. 2. Say: 'Statisticians have names for these ideas. We are going to meet them properly in the coming lessons. Today I just want you to know they exist.' Write on the board: FREQUENCY TABLE | MEAN / MEDIAN / MODE | HISTOGRAM | FREQUENCY POLYGON. Ask: 'Has anyone heard any of these words before? Where?' [Quick show of hands and 2 cold-calls — do not explain the terms yet.] 3. Teacher models the rough histogram on the board in under 3 minutes — narrate aloud: 'I am putting prices 86 to 91 in the first group... 92 to 97 in the second group...' Draw bars. Ask: 'What do	Concept Introduction through Naming and Previewing (not yet teaching procedures): by seeing the rough histogram emerge from chaos on the board in real time, learners experience the 'aha' of visual pattern recognition before they have the formal skill to create it themselves. Naming the tools (frequency table, mean, median, mode, histogram) without defining them creates schema anchors — learners will attach meaning to these words as the sub-strand progresses. This is analogous to the NGSS practice of 'Analysing and Interpreting Data' at a surface level before 'Using Mathematics and Computational Thinking.'	Observable indicators: (i) Learners can articulate in their own words why the raw list is insufficient — listen for language like 'we can't see the pattern,' 'we don't know what is normal,' 'we need to sort it first'; (ii) At least two-thirds of the class responds to the histogram reveal with visible surprise or engagement (leaning forward, spontaneous commentary) — this is the desired 'aha' response; (iii) Exit check: ask learners to write one sentence completing: 'Before we can use the Wakulima Market data to settle the argument, we need to...' Scan 6–8 responses before moving to Phase 4.

	most prices cluster in the middle bands. Teacher asks: 'Looking at this picture — who was more right, the family or the neighbour?' This is revisited but not yet fully resolved — learners are told they will build the proper tools to answer it definitively over the coming lessons.	you notice about where most prices fall?' [WAIT 10 seconds. Cold-call 2 learners.] 4. Ask the key question: 'Does this picture help settle the argument — or does it make things more complicated?' [WAIT 12 seconds. Expect productive disagreement.] 5. Close by naming the tension: 'We have not settled the argument yet — and that is deliberate. We now know what tools we need. The next lessons will give us those tools properly, and then we will come back and settle this mystery for good.'		
Phase 4 — Driving Question Board (DQB) Creation 📹 **VIDEO:** Statistics: Alternate variance formulas Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher introduces the Driving Question Board (DQB) — a dedicated space on the classroom wall (or a large sheet of manila paper) that will stay up for the entire sub-strand. Learners are given two sticky notes each (or small paper squares if sticky notes are unavailable). On the first note, they write one question they now have about the Wakulima Market data or about statistics in general that they want answered by the end of the sub-strand. On the second note, they write one thing they noticed or found surprising during today's lesson. Learners post their notes on the DQB, grouping similar questions together as they do so. The teacher reads a selection aloud, organises them into clusters (e.g., 'Questions about averages,' 'Questions about graphs,' 'Questions about what makes data reliable'), and labels the clusters. The three KICD Key Inquiry Questions are written in a different colour at the top of the DQB as the overarching frame. The DQB will be revisited and updated at the	1. Introduce the DQB: 'This board will be the memory of our learning journey. Every question you have — every confusion, every curiosity — goes here. By the time we finish this sub-strand, every cluster on this board should have an answer.' 2. Distribute two notes per learner. Give 3 minutes of silent writing. Narrate: 'First note — your best question. It can start with How, Why, What, or Can. Second note — one surprising thing from today.' 3. Invite learners to come up and post notes. While they do, teacher reads 3–4 aloud and asks: 'Does anyone else have a question like this one?' — use this to physically cluster similar notes. 4. Name the clusters aloud: 'I can see we have a cluster of questions about how to find the middle price — we will answer those in Lesson 3. A cluster about what the graph should look like — Lesson 5. A cluster about whether statistics can lie — that is one of the most important questions in this sub-strand.' 5. Point to the three KICD Key Inquiry Questions written at the top: 'Notice that your questions	Driving Question Board as a Living Artefact: the DQB externalises learners' metacognitive awareness — they can see their own curiosity organised spatially. Clustering questions into categories previews the structure of the sub-strand (collection → organisation → measures of centre → representation → interpretation) without the teacher imposing that structure top-down. This builds learner agency and a sense of ownership over the inquiry. Revisiting and updating the DQB each lesson creates a visible record of growing understanding, which is motivating and reinforces the sub-strand's storyline.	Observable indicators: (i) Every learner posts at least one question — a blank note indicates disengagement or anxiety; approach such learners quietly and ask 'What confused you most today?' to generate a question; (ii) The quality of questions reveals depth of engagement — surface questions ('What is mean?') are expected and valid; deeper questions ('Can two different sets of data have the same average but look completely different?') indicate high prior knowledge and can be used to stretch those learners; (iii) Teacher photographs or lists the questions for planning purposes — questions that appear frequently become priority teaching targets in subsequent lessons.

	end of every lesson in the sub-strand.	are versions of these three big questions. Everything we do connects back to them — and to our Wakulima Market mystery.' 6. Take a photo of the DQB (if a camera is available) so the initial state is recorded for comparison at the end of the sub-strand.		
Phase 5 — Model Building 📹 **VIDEO:** [Statistics: Alternate variance formulas] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Summary Statistics, Summarizing Univariate Distributions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	For this opening lesson, the 'model' is a conceptual map rather than a mathematical model. Learners are given a half-page template with a central box labelled 'The Wakulima Market Maize Price Data (40 values)' and four empty branches labelled: (1) What we know about this data already; (2) What is confusing or unclear; (3) What tools we think we need; (4) What we predict the final answer will be. Working individually for 5 minutes, learners fill in the branches using words, phrases, or simple sketches. This becomes their personal baseline model — a record of where their understanding starts. Learners store this in their exercise books on a dedicated 'Model' page that will be revised after every lesson. The teacher closes by returning to the phenomenon: 'The family and the neighbour are both waiting for our answer. We are not ready to give it yet — but today we know exactly what we need to learn to get there. That is what it means to think like a statistician.'	1. Distribute the template or instruct learners to draw the four-branch map in their exercise books. Say: 'This is your starting model. It does not need to be right — it needs to be honest. Write what you actually think, not what you think I want to hear.' [WAIT 5 minutes of individual silent work — do not assist during this time.] 2. After 5 minutes, ask 3 learners to share one branch each (cold-call: 'Wanjiru, read us your branch 3 — what tools do you think we need?'). Do not correct or refine — simply affirm: 'That is your starting point. By the end of this sub-strand, you will come back to this page and see how your thinking has grown.' 3. Instruct learners to write on top of the page: 'Lesson 1 — Before learning.' They will add 'Lesson X — After learning' updates on subsequent pages. 4. Close with the phenomenon: 'Next lesson, we will begin building the first real tool — the frequency distribution table. That tool is going to start giving us actual answers about those 40 prices. The mystery continues.' 5. Optional extension for fast finishers: 'If you go to a market this week — Wakulima, Gikomba, Toi, or your local market — notice the price of unga or any staple food. Write it down and bring it next lesson. We may add it to our data set.'	Baseline Conceptual Mapping: asking learners to construct a four-branch conceptual map before instruction begins serves as both a formative pre-assessment and a metacognitive anchor. When learners revisit and revise this map in later lessons, they can literally see their understanding growing — this is a powerful motivator and supports self-regulated learning. The model is explicitly framed as a living document, not a test, which reduces anxiety and encourages honest self-assessment. This mirrors the NGSS practice of 'Developing and Using Models' at its earliest, most intuitive stage.	Observable indicators: (i) Branch 1 ('What we know') — learners should be able to identify at least: there are 40 prices, they are in KES, they come from Wakulima Market, the range is approximately 86–115 KES; if a learner's branch 1 is completely empty, they need additional orientation to the data; (ii) Branch 3 ('Tools we need') — listen for words like 'average,' 'table,' 'graph,' 'sort' — these indicate readiness; (iii) Branch 4 ('Prediction') — learners who predict a specific answer (e.g., 'The neighbour is more right because most prices are in the middle') are demonstrating early inferential reasoning — note these learners for extension opportunities; (iv) Collect 5–6 maps at random to read before Lesson 2 — use them to calibrate the starting point of the class.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions before planning Lesson 2: 1. PHENOMENON ENGAGEMENT: Did the Wakulima Market scenario feel real and urgent to learners, or did it feel like a contrived word problem? What adjustments to the framing might make it more immediate? 2. PRODUCTIVE STRUGGLE: During Phase 2, did learners experience genuine frustration with the raw data — or did they give up too quickly / find it too easy? If groups gave up, consider whether the data sheet needs to be more visually chaotic. If groups found it too easy (by spontaneously sorting), accelerate to frequency tables earlier in Lesson 2. 3. DQB QUALITY: Read through the DQB questions. How many are procedural ('How do you calculate mean?') versus conceptual ('Why would two summaries of the same data give different answers?')? A class dominated by procedural questions may need more phenomena-driven instruction; a class with rich conceptual questions can move faster. 4. PREDICTION SPLIT: Was the class split roughly even between 'Team Family' and 'Team Neighbour'? A very uneven split (e.g., 90% one way) may indicate the scenario needs rebalancing. Note which prediction the class majority made — this will create a satisfying resolution moment when the data is fully analysed in a later lesson. 5. KENYA CONTEXT: Did learners connect personally to the Wakulima Market context? If the school is outside Nairobi, consider adapting the scenario to the local market (e.g., Kongowea Market in Mombasa, Kibuye Market in Kisumu, or the local open-air market). The data set remains the same; only the market name changes. 6. INCLUSION: Were all learners able to access the task? Note any learners who struggled with reading the raw data list or who did not post a DQB question. Plan targeted support for Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	A chaotic list of 40 daily maize flour prices from a Wakulima Market stall, collected over 8 weeks. A family says prices vary wildly; their neighbour says prices are always about the same. Both shop at the same stall.
What did I learn?	Raw, unorganised data cannot settle a real-life disagreement fairly. You can find the minimum and maximum price by scanning the list, but you cannot easily see patterns, typical values, or the overall shape of the data. To use data honestly and usefully, you need tools to organise and represent it.
How does this explain the phenomenon?	Statistics is the science of collecting, organising, representing, and interpreting data to support honest decision-making. The Wakulima Market mystery cannot be solved by looking at raw numbers alone — we need frequency tables, measures of central tendency (mean, median, mode), and graphs (histograms, frequency polygons). These are the tools we will build in the coming lessons to finally settle the argument between the family and their neighbour.

LESSON 2 (80 minutes): Meeting the Maize Price Data: What Is Raw Data and Where Does It Come From?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners encounter the Wakulima Market maize price dataset for the first time, discover the limits of raw data, and begin building the vocabulary and habits of mind needed to organise and interpret data throughout the unit.
Knowledge	By the end of this lesson the learner should be able to: (a) collect data from real-life sources; (b) explain the meaning of the terms data, raw data, data sources, tally, and frequency; (c) distinguish between primary and secondary data sources.
Skills	Learners practise: recording data systematically using tally marks; posing statistical questions; collaboratively constructing a Driving Question Board entry; beginning collection of a class data set (time taken to walk to school) for use throughout the unit.
Attitudes	Integrity — the learner collects and records data accurately. Unity — the learner works in groups to engage with the shared class data set. Love — the learner is considerate of peers as they discuss conflicting interpretations of the same raw data.
Key Inquiry Question	What is statistics? How do we use statistics in day-to-day life?
Purpose in Storyline	This is Lesson 2 of 8 in the evidence-gathering sequence. In Lesson 1 learners were introduced to the anchoring phenomenon (the Wakulima Market maize price mystery) and made initial predictions. In this lesson they handle the raw data list for the first time. The cognitive conflict between the family ('prices vary wildly') and the neighbour ('prices are always about the same') is kept deliberately unresolved — learners discover that staring at 40 unorganised numbers cannot settle the argument, creating an urgent felt need for data organisation tools introduced in subsequent lessons. Learners also begin collecting their own class data set (walk-to-school times) that will be used in every remaining lesson.
Safety Notes	No physical hazards. Ensure learners do not share personal home-address information when discussing walk-to-school distances. Emotional safety: affirm that all predictions are valid starting points — no prediction is 'wrong' at this stage.

B. LESSON OVERVIEW
Learners are shown a handwritten list of 40 daily maize flour prices (in Kenyan shillings per 2 kg packet) collected over eight weeks from a stall at Wakulima Market, Nairobi. Working first individually, then in pairs, they attempt to answer the driving question — 'Who is right, the family or the neighbour?' — using only the raw list. They quickly discover this is nearly impossible without organisation. The class unpacks key vocabulary (data, raw data, data sources, tally, frequency) through a structured word-wall activity. Learners then collect a real class data set — the number of minutes each learner takes to walk to school — record it using tally marks, and post the first entry on the class Driving Question Board. The lesson closes with learners articulating what further tools they will need to answer the maize price question.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
1. Predict Phase (15 minutes)	Each learner receives a printed or projected copy of the 40 raw maize prices (e.g., 98, 112, 105, 99, 130, 104, 97, 118, 101, 110, 125, 103,	Display the 40 prices prominently on the board or screen. Say exactly: 'You have 90 seconds — no talking, just look at these	Think-Pair-Share with recorded predictions. This activates prior intuitive experience with market prices (virtually all Kenyan learners	Observable indicator: every learner has a written sentence in their book before pair-sharing begins. Teacher circulates during

📹 **VIDEO:** Statistics: Alternate variance formulas Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	99, 107, 122, 98, 115, 104, 100, 130, 96, 108, 121, 103, 99, 114, 106, 127, 102, 110, 98, 105, 119, 104, 96, 112, 107, 128, 101, 109 — all in KSh). Learners are given exactly 90 seconds of silent individual thinking time to scan the list and write one sentence in their exercise books: 'I think the family / the neighbour is more right because…'. They then share their sentence with their seat partner (Think-Pair). No consensus is required at this stage.	numbers and write one sentence in your book. Is the family right that prices vary a lot, or is the neighbour right that they are stable? Begin.' [Set a visible 90-second timer.] After the timer: 'Share your sentence with the person next to you. You have 60 seconds.' After sharing: cold-call 4 learners — select a mix of those who chose 'family' and those who chose 'neighbour'. Ask each: 'What in the list made you think that?' [Apply 12-second wait time before cold-calling if no volunteer responds.] Record the two positions ('FAMILY — varies' and 'NEIGHBOUR — stable') as column headers on the board. Place tally marks under each as learners share. Do NOT resolve the conflict — say: 'We cannot agree yet. That is the point. Let us find out why.'	have accompanied a family member to a market) and externalises the two competing claims so the cognitive conflict is visible to the whole class. Writing the prediction first prevents anchoring to the first peer response heard.	the 90 seconds and notes (without correcting) whether learners are looking for the highest and lowest values, scanning for repeated values, or expressing confusion — this informs the depth of vocabulary instruction needed in Phase 3. Red flag: if more than half the class says 'I cannot tell' — this confirms the lesson's central insight is already being felt and should be named explicitly.
2. Observe Phase (20 minutes) 📹 **VIDEO:** Statistics: Alternate variance formulas Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs. Each pair receives a task card with three questions: (Q1) Without doing any calculations, find the highest and lowest price in the list. (Q2) Try to find the price that appears most often — just by looking. (Q3) Write one sentence: Is this list easy or hard to work with? Why? Pairs have 10 minutes to work on the task card. After 10 minutes, the class transitions to the second observation activity: the teacher leads a whole-class live data collection. Every learner stands, and one at a time (or by show of hands in bands), reports their walk-to-school time in minutes. The class secretary (a rotating student role) records each response on the board in real time as a running list — no organisation yet. This second raw list becomes	Circulate during pair work. Listen for the language learners use — do they say 'numbers', 'figures', 'prices', or 'data'? Note this for vocabulary instruction. After 10 minutes, cold-call 3 pairs to share Q3 answers. Ask: 'What made it hard?' [12-second wait time.] Affirm: 'Everyone is finding this hard. That frustration has a name in mathematics — and we will name it in a moment.' For the live data collection: say 'Stand up if your walk to school takes 0–10 minutes.' Count and record. Repeat for 11–20, 21–30, 31–40, 41+ minutes. Then say: 'We now have two raw data lists on our board — the Wakulima maize prices, and our own walk-to-school times. Both are raw. Both are hard to read. Remember this feeling.'	Parallel data sets — one external (Wakulima prices) and one personal (walk-to-school times). Using a personal data set dramatically increases engagement and ownership. The deliberate decision NOT to organise the data during this phase preserves the productive frustration that motivates the vocabulary and organisation tools introduced next. Connecting the two data sets establishes that the skills learned on the maize data apply to their own lives.	Observable indicator: pairs can identify the maximum (KSh 130) and minimum (KSh 96) from the raw list within 10 minutes — this confirms basic data-reading ability. Observable indicator: at least one learner per pair articulates that finding the most frequent value 'by eye' is unreliable — this is the key insight of the phase. Teacher notes which pairs attempt to sort or group the data informally — these learners are ready for extension in later lessons.

	the class's own data set for the rest of the unit.			
3. Explain Phase (20 minutes) 📹 **VIDEO:** Statistics: Alternate variance formulas Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher introduces a Word Wall with five terms: DATA, RAW DATA, DATA SOURCE, TALLY, FREQUENCY. For each term, the teacher gives a one-sentence KICD-aligned definition, then immediately anchors it to either the Wakulima list or the walk-to-school list. Learners then practise using tally marks: they are given the walk-to-school raw list from the board and in pairs they organise it into a simple tally table (bands: 0–10 min, 11–20 min, 21–30 min, 31–40 min, 41+ min) and determine the frequency for each band. Learners record the completed tally table in their exercise books. The teacher then returns to the maize price list: 'Now that we know what raw data is, why was it so hard to answer the family-vs-neighbour question using just this list?'	Use a Word Wall card for each term — post each card as it is introduced. For DATA: 'The 40 maize prices on the board — each price is a piece of data. Our walk-to-school times are also data.' For RAW DATA: 'Before we organise it — like this jumbled list — we call it raw data. Like raw maize before it is milled into unga.' [The unga analogy is culturally resonant and memorable.] For DATA SOURCE: 'The stall-keeper at Wakulima Market is a primary data source. A newspaper reporting average maize prices is a secondary source. Which did we use for our walk-to-school data?' [Cold-call 2 learners, 10-second wait time.] For TALLY and FREQUENCY: model one row of the tally table live on the board, then release pairs to complete the rest. After pairs complete the tally table, cold-call 3 pairs to share their frequency for one band — write on the board and check for consensus. Close the phase by returning to the phenomenon: ask 'Now that you know what raw data is — why could neither the family nor the neighbour give us a clear answer just from the raw list?' [Cold-call 2 learners, 12-second wait time.] Write the best answer verbatim on the board.	Concrete-Representational-Abstract (CRA) anchored to phenomenon. The unga analogy (raw maize → milled flour) gives learners a culturally grounded mental model for raw vs. organised data. Immediate application of each vocabulary term to both the phenomenon data set and the personal data set ensures terms are not abstract definitions but tools for a real task. The tally table is the first data organisation tool in the unit — its simplicity is deliberate so learners feel competent before complexity increases.	Observable indicator: every learner has a completed tally table in their exercise book with correct frequencies summing to the class size. Teacher spot-checks 5–6 books during pair work. Key misconception to watch: learners using tally marks in groups of 4 rather than groups of 5 (IIII not ||||) — correct immediately with a brief whole-class pause. Observable indicator: at least one learner can articulate that raw data hides the distribution — e.g., 'You cannot see which price appears most because the numbers are not grouped.'
4. Driving Question Board (DQB) Creation (10 minutes) 📹 **VIDEO:** Statistics: Alternate	Each learner writes two sticky notes (or index cards): one ORANGE card — something they now understand about the maize price mystery that they did not understand before this lesson; one BLUE card — a new question the lesson has raised for them.	Say: 'You have 3 minutes — write your orange card first (what you now understand), then your blue card (what you still want to know). Use complete sentences.' Circulate and prompt any learner who is stuck with: 'What was the most surprising thing about that	Two-colour DQB protocol (understanding + new questions). The orange/blue distinction prevents the DQB from becoming a pure question list — it also tracks growing understanding, making learning progress visible. Reading cards aloud without attribution	Observable indicator: every learner posts at least one card of each colour. Teacher scans orange cards for evidence that the core insight (raw data is hard to interpret) has landed — if fewer than 60% of orange cards reflect this, the teacher should open the

variance formulas Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners post their cards in the two labelled columns of the class DQB. The teacher reads 3–4 cards aloud and groups similar questions visibly on the board. The class agrees on one 'Most Burning Question' to add as a formal DQB entry — typically something like: 'How can we organise the 40 prices so we can actually see whether they vary a lot or a little?'	raw data list?' After posting, read 3 orange and 3 blue cards aloud without attribution. Group blue cards that ask similar questions by physically moving them together on the DQB. Ask the class: 'Which of these questions do you most want us to answer in the next lesson?' Take a show of hands between the top 2 candidate questions. Write the winning question on a large DQB card in a different colour from previous lessons' entries. Point to the growing DQB: 'This board is our evidence trail. Every lesson, we add to it. By Lesson 8 it will tell the full story of the maize price mystery.'	creates psychological safety and models academic language for learners who struggle to articulate mathematical thinking in English. The class vote gives learners agency over the inquiry direction and increases motivation for the next lesson.	next lesson with a brief re-visit. Blue cards serve as a formative pre-assessment of readiness for frequency distribution tables (Lesson 3) — questions about 'grouping' or 'sorting' indicate readiness; questions still focused on 'which price is correct' indicate a need to reinforce data vocabulary.
5. Model Building Phase (15 minutes) 📹 **VIDEO:** Statistics: Alternate variance formulas Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the cumulative class model begun in Lesson 1 — a large annotated diagram on poster paper titled 'How We Will Solve the Maize Price Mystery'. In Lesson 1, learners drew the phenomenon (the family and the neighbour at the market) and wrote their initial prediction. In this lesson, each group adds two things to the model: (1) a label on the raw data list in the diagram — they write 'RAW DATA — hard to interpret' with an arrow pointing to the jumbled number list; (2) a small tally table (using their walk-to-school data) glued or drawn in a corner of the poster, labelled 'TALLY TABLE — first step of organisation'. Groups share one addition to the model with the class in a 30-second gallery-walk format (groups rotate one position and read the neighbouring group's additions).	Distribute the Lesson 1 group posters. Say: 'Your model is a living document. Today you are adding two new pieces of evidence to it. You have 8 minutes.' Circulate and ask each group: 'What does your tally table tell you that the raw list did not?' [10-second wait time before prompting further.] After 8 minutes: 'Rotate one group to the right. Read your neighbours' additions. You have 90 seconds.' After the gallery walk, cold-call 2 learners to share one thing they noticed on a neighbouring group's poster. Close with: 'In Lesson 1 we asked: who is right — the family or the neighbour? Look at your model. Can we answer that yet?' [Expected answer: No — we only have raw data and a simple tally. We need more tools.] Say: 'That is exactly where we are. The mystery is still open. Next lesson, we get our first real tool for solving it.' Post all group models on the wall for ongoing reference.	Cumulative model revision. The physical poster model makes the incremental nature of scientific and mathematical inquiry visible and concrete. Adding a tally table to the model alongside the raw data list creates a side-by-side comparison that reinforces the lesson's central insight. The gallery walk activates peer learning without requiring formal presentations, keeping energy high at the end of the lesson. Ending on an explicit statement of what is still unknown ('the mystery is still open') sustains intellectual curiosity and primes motivation for Lesson 3.	Observable indicator: every group poster contains both required additions (raw data label and tally table) before the gallery walk begins. Teacher photographs each updated poster for use in Lesson 3. Key quality indicator: the tally table on the poster uses correct tally-mark groupings (groups of five) and frequencies that match the class data collected in Phase 2. Observable indicator during gallery walk: learners can read a neighbouring group's tally table and state the modal band (the band with the highest frequency) — this previews the concept of mode introduced in Lesson 4.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions in your professional journal: (1) Did the 'raw unga' analogy for raw data resonate with learners — did you hear them reusing it during pair work or the model-building phase? If not, what local analogy might work better for your class? (2) Were learners genuinely frustrated by the raw data list, or did the frustration feel forced? Genuine frustration is the engine of this unit — if it was not present, consider making the raw list even messier in presentation (handwritten, out of order, varying font sizes) next time. (3) How many blue DQB cards asked specifically about 'grouping' or 'organising' the data? If fewer than half, plan an opening hook for Lesson 3 that makes the need for a frequency distribution table feel even more urgent — for example, ask two learners to race: one finds the most common price from the raw list, one from an organised table. (4) Was your cold-call distribution equitable across gender, seating zones, and perceived ability? Review your notes and ensure no group is systematically silent across lessons. (5) The walk-to-school data set belongs to this class and will be used for every remaining lesson — store it safely and confirm that every learner has the full class data set recorded in their exercise book before they leave today.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners examined a raw list of 40 maize flour prices from Wakulima Market and found it nearly impossible to determine — by eye alone — whether prices were stable or variable. They also observed a live raw data set of walk-to-school times collected from their own class, which presented the same challenge of unorganised numbers.
What did I learn?	Raw data is data that has not yet been organised or summarised. Data can come from primary sources (collected directly, e.g., asking classmates) or secondary sources (e.g., a market price list). Tally marks are a tool for organising raw data into frequencies. A frequency is the count of how many times a value or group of values appears in a data set.
How does this explain the phenomenon?	The reason both the Nairobi family and their neighbour could be telling the truth is that raw data — a jumbled list of numbers — does not reveal patterns clearly. Without organising the 40 maize prices, different people notice different numbers and reach different conclusions. To settle the argument fairly, we need tools that organise the data so the full picture becomes visible.

LESSON 3 (80 minutes): Organising the Chaos: Frequency Distribution Tables for Ungrouped Data	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners construct frequency distribution tables for ungrouped data drawn from the Wakulima Market maize price dataset, enabling them to begin transforming a chaotic raw list into an organised structure that reveals patterns — a critical first step toward settling the family-vs-neighbour dispute.
Knowledge	By the end of this lesson, learners should be able to: (a) explain what a frequency distribution table is and identify its key components (data value, tally, frequency); (b) construct a frequency distribution table for ungrouped data from a real-life dataset; (c) identify the modal value directly from a completed frequency distribution table.
Skills	Tallying and organising raw data systematically; constructing a clearly labelled frequency distribution table; reading and interpreting frequency counts; identifying patterns in organised data that are invisible in raw form.
Attitudes	Integrity — recording tally marks and frequencies honestly without adjusting data to fit a preferred answer; Unity — collaborating in pairs and groups to share tallying workload fairly; Love — being considerate of peers' reasoning as groups compare and reconcile their tables.
Key Inquiry Question	How do we represent data?
Purpose in Storyline	In Lesson 2 learners experienced the anchoring phenomenon and explored why the raw list of 40 maize prices feels chaotic — they could not easily answer who was right, the family or the neighbour. In Lesson 3 learners take the critical next step: they organise that same raw data into a frequency distribution table for ungrouped data. For the first time the data stops being a jumble of numbers and becomes a structured picture. Learners will see which prices appear most often (the mode), which appear rarely, and whether the distribution is tight or spread — giving them their first real evidence to begin answering the driving question honestly.
Safety Notes	No physical hazards in this lesson. Remind learners to use rulers carefully when drawing table lines. Ensure data worksheets are shared equitably so no group is left without materials.

B. LESSON OVERVIEW

Learners work with the 40 Wakulima Market daily maize flour prices introduced in the anchoring phenomenon. Beginning with a brief predict phase (Can organising the data change what we see?), learners move into a structured observe phase in which pairs tally the raw prices using a class-constructed tally chart template. In the explain phase, pairs combine their tallies into a whole-class frequency distribution table on the board, then use the completed table to answer three targeted questions about patterns in the data. The DQB is updated with new evidence-based claims and refined questions. Finally, learners update their cumulative data model by sketching the shape of the distribution as suggested by the frequency table, building a conceptual bridge toward histograms in the next lesson. Throughout, all work is anchored to the question: does this organised table help us decide whether the Nairobi family or their neighbour is telling the truth about maize prices at Wakulima Market?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 —	Each learner receives a half-page	Display the raw price list	Predict-then-Compare (Anchored	Observable indicator: Every

Predict 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Line Plots from Frequency Tables (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	card showing the same chaotic raw list of 40 daily maize flour prices (in Kenyan shillings) that was introduced in the anchoring phenomenon — e.g. prices ranging roughly between KSh 90 and KSh 150 per 2 kg packet, with many repeated values. Learners individually write answers to two predict prompts on a sticky note or exercise book: (1) 'If you organised these 40 prices into a neat table, do you think the price the family experienced most often would become clearer? YES / NO — explain your thinking in one sentence.' (2) 'Predict: roughly how many different price values do you think exist in this list of 40 prices? Write a number.' Learners hold their sticky notes but do not share yet. The teacher cold-calls 5 learners to read their predictions aloud before any instruction, recording predictions on the side of the board under the heading 'Our Predictions — Before Organising'. The teacher does NOT confirm or correct — only thanks each learner and records. This creates a public prediction record the class will return to.	prominently (printed worksheet or projected). Say verbatim: 'You have 3 minutes — read the list, then answer both prompts on your own. Do not discuss yet.' Set a visible 3-minute timer. After time, say: 'I am going to cold-call five people to share prediction number one. I want the honest prediction you wrote — there is no wrong answer right now.' Cold-call 5 learners (aim for gender balance and mix of confidence levels). After each response, wait 10 seconds silently before calling the next. Record each prediction verbatim on the board. Then ask the whole class: 'Raise your hand if your prediction for the number of different prices was less than 10.' Count and note. 'Between 10 and 20?' Count and note. 'More than 20?' Count and note. Say: 'Let us find out together whether organising this data changes what we can see — and whether it helps us settle the argument at Wakulima Market.'	Prediction): By making a public, recorded prediction before any instruction, learners activate prior knowledge about organisation and pattern, and create personal cognitive commitment. When evidence later confirms or challenges their prediction, the contrast drives deeper encoding. Recording predictions on the board makes them a shared class artefact that is revisited explicitly at the end of the lesson.	learner has written a YES/NO response with at least one sentence of reasoning AND a numerical estimate. The teacher scans the room during the 3 minutes to confirm all learners are writing (not blank pages). The range of cold-called predictions reveals the class's baseline intuition about data organisation — if most learners say 'NO, organising won't help', the teacher notes this as a productive misconception to address in the explain phase.
Phase 2 — Observe 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Line Plots from Frequency Tables (Flexbook)] Source: CK-12	Learners work in assigned pairs. Each pair receives: (a) the same raw list of 40 maize prices; (b) a blank three-column tally table template pre-drawn on paper with headers: 'Price (KSh) | Tally | Frequency'. The teacher explains the task: pairs must go through the raw list one price at a time, find that price value in their tally column (or add a new row if it is new), and mark one tally stroke. No calculators — only tally marks and counting. Pairs have 12 minutes. At the 6-minute mark the	Before pairs begin, model the first three entries of the tally table using a think-aloud: 'I look at the first price — KSh 110. I find row 110 in my table and make one tally stroke like this. Next price — KSh 125. New row, one tally stroke. Next price — KSh 110 again. I go back to the KSh 110 row and add a second tally stroke.' Do not complete the table — stop after three entries and say: 'Now you and your partner continue for all 40 prices. Divide the list — one partner reads prices aloud, the	Hands-On Data Organisation (NGSS Science and Engineering Practice 4 — Analysing and Interpreting Data): Learners physically handle the raw data and impose structure on it through tallying. This embodied process makes the concept of frequency concrete — frequency is not an abstract number but a count of real events (real prices paid by real Nairobi families). The mid-task pause and the sum-to-40 check build in self-monitoring habits that are foundational to data integrity.	Observable indicator: Each pair produces a completed tally-and-frequency table whose frequency column sums to 40. The teacher checks at least 6 pairs' tables during circulation. Key error to watch for: duplicate rows for the same price (e.g. two rows for KSh 120) — this indicates the pair did not organise by value first. Key success indicator: pairs spontaneously begin to notice which prices have tall tally columns before being asked.

🔍 Search ARES for similar readings	teacher calls a 'mid-task pause': each pair counts how many distinct price values they have found so far and holds up fingers. After 12 minutes, pairs convert tally marks to frequency counts and write the frequency number in the third column. Each pair then totals the frequency column and checks it sums to 40. Pairs that finish early are asked: 'Look at your completed table. Which price appears most often? Circle it. What do we call that value in statistics?'	other marks tallies. Switch halfway.' Set a 12-minute timer. Circulate actively: (1) Check that pairs are working through the list in order, not skipping; (2) At the 6-minute mid-pause, cold-call 3 pairs: 'How many different price values have you found so far?' Record these on the board. (3) After 12 minutes, say: 'Convert tallies to frequencies now — 2 minutes.' Then ask: 'Everyone check your frequency column sums to 40. Raise your hand when you have confirmed this.' Wait 15 seconds. For any pair whose total ≠ 40, ask: 'Which row might have a tally error? Check rows with five-bar groups first.' Do NOT give the answer — prompt them to self-correct.		
Phase 3 — Explain 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Line Plots from Frequency Tables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class constructs a single agreed frequency distribution table on the board. The teacher writes the column headers and calls on different pairs to contribute one row each — each pair reads out one price value and its frequency while the teacher (or a student scribe) fills in the row. Once the full table is on the board, learners copy it into their exercise books. The teacher then poses three structured explain questions, allowing think-pair-share for each: (Q1) 'Looking at this table — which single price appears most often? What is the statistical name for this value?' (Q2) 'The family said prices vary wildly. The neighbour said prices are stable. Does this organised table give you any evidence yet for either side? Explain using a specific row from the table.' (Q3) 'Our table has frequencies but we still have to read every row. What would make	For the board table construction: assign each pair a specific price range to contribute (e.g. 'Pair 1 — give me all prices between KSh 90 and KSh 99'). This prevents repetition and keeps all pairs engaged. After the full table is built, ask: 'Everyone — does your pair's table match ours? If a row is different, raise your hand.' Resolve discrepancies by going back to the raw data publicly — model data integrity. For Q1: after 10 seconds of wait time, cold-call 3 learners. If a learner says 'the most common price' without using the word 'mode', ask: 'Does anyone know the mathematical term for that?' For Q2: after think-pair-share, cold-call one pair that predicted YES (the family is right) and one that predicted the neighbour is right, so both positions are heard. Say: 'We are not settling this today — we are collecting evidence. Hold your claim.' For Q3: after	Structured Argumentation from Evidence (NGSS Science and Engineering Practice 7 — Engaging in Argument from Evidence): Q2 explicitly requires learners to make a claim and cite a specific row of the table as evidence. This is the core epistemic move of statistics — using organised data to support or challenge a claim. By hearing both the 'family is right' and 'neighbour is right' arguments from the same table, learners experience productive cognitive conflict: the table can support both claims depending on which rows you focus on. This motivates the need for summary statistics (mean, median, mode) and graphical displays in subsequent lessons.	Observable indicator: During cold-calls on Q2, learners cite a specific price and frequency (e.g. 'KSh 120 appears 8 times, which means it is quite common — so the neighbour might be right that prices cluster around that value'). If learners give only vague answers ('the numbers are close together'), the teacher prompts: 'Which row in our table shows you that? Read it out.' Exit-ticket equivalent: each learner writes the completed frequency distribution table in their book AND circles the mode — the teacher checks these during the DQB phase.

	it even easier to see at a glance whether most prices are clustered together or spread far apart?' (Q3 is deliberately open-ended and forward-pointing — it seeds the need for a histogram.) After think-pair-share on each question, the teacher cold-calls 3 pairs per question. The class then revisits the public predictions recorded in Phase 1: which predictions were confirmed? Which were wrong? Learners update their sticky notes with one sentence beginning 'I now think…'	cold-calling 3 pairs, say: 'Excellent. You have just described why we need a histogram. That is our next lesson.' This explicit bridging statement connects the current lesson to the broader sequence. Return to the prediction board: strike through predictions that the evidence has now addressed, and circle those that remain open.		
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Line Plots from Frequency Tables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board displayed on the classroom wall. The DQB has three columns: 'What We Have Observed', 'What We Now Know/Claim', 'What We Still Wonder'. In pairs, learners write on two sticky notes: (Note 1 — green) one specific thing they can now claim about the Wakulima Market maize prices based on the frequency distribution table, citing at least one piece of evidence (e.g. a specific frequency). (Note 2 — orange) one new question the frequency table raised for them that it cannot yet answer. Pairs come up and place their notes in the correct columns. The teacher reads 4–5 notes aloud and facilitates a brief 5-minute whole-class discussion to identify: (a) the most common new claim, (b) the most interesting new question. The teacher explicitly names two questions that will be answered in upcoming lessons: 'Several of you are asking — what is the average price? That is the mean and median — Lesson 4. Several of you are asking — can we draw a picture of this table? That is the	Before pairs write, show the DQB and remind them: 'A claim must be tied to evidence from our table. Do not write feelings — write data.' Give pairs 4 minutes to write their two notes. As they post notes, group similar claims together visually (you can do this in real time or ask a student helper). When reading notes aloud, read exactly what is written — do not paraphrase, to honour learner voice. After identifying the top claim and question, say: 'Our Driving Question asks how we use statistics to tell an honest story. Does our frequency table tell an honest story yet? What is it still hiding?' Wait 10 seconds. Cold-call 3 learners. This question primes the need for graphical displays and measures of central tendency without giving away the answer.	Driving Question Board as Cumulative Evidence Tracker (Phenomenon-Anchored Inquiry): The DQB makes the class's growing understanding visible and public. By requiring evidence-based claims (not opinions), it reinforces the epistemic norm that statistics is about what the data shows, not what we feel. The explicit teacher move of naming which upcoming lessons will answer specific DQB questions gives learners a sense of purposeful progression — they are not doing isolated exercises but building a coherent argument across 8 lessons.	Observable indicator: Green sticky notes contain a specific numerical claim linked to the frequency table (e.g. 'KSh 120 is the modal price — it appears X times'). Orange sticky notes ask questions that go beyond what the table shows (e.g. 'But what is the average price?', 'Why do some prices appear so rarely?'). If green notes are vague ('the data is organised now'), the teacher pauses the class and models a strong evidence-based claim using a row from the board table before pairs finalise their notes.

	histogram — Lesson 5.'			
Phase 5 — Model Building 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Line Plots from Frequency Tables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner opens to their 'Cumulative Data Model' page — a dedicated page in their exercise book where they have been building a model of the maize price phenomenon since Lesson 1. Today they add Layer 3: a hand-drawn dot-plot or frequency bar sketch derived directly from the completed frequency distribution table. The instruction is: 'On the x-axis write the different price values from your table. Above each price, draw a column of dots (or a bar) whose height equals the frequency for that price. You do not need rulers or perfect scale — this is a sketch to help you see the shape.' Learners have 6 minutes to complete the sketch. They then answer one final prompt in their model journal: 'In one sentence, describe the shape you see — are most prices clustered in one area, or are they spread out? How does this shape begin to answer our Driving Question?' Learners share their shape descriptions with a shoulder partner before the lesson closes.	Display a partially drawn example dot-plot on the board to model what is expected — draw 3 price values with their dot columns, then stop and say: 'Continue this for all values in your table. The shape that emerges is called the distribution. Remember it — we will return to it when we draw histograms.' As learners sketch (6 minutes), circulate and look for: (1) learners who draw bars of equal height regardless of frequency — redirect them back to the table; (2) learners who skip rare prices (frequencies of 1 or 2) — remind them that every row in the table must appear on the x-axis. In the final 2 minutes, cold-call 3 learners to read their one-sentence shape description aloud. After each, ask the class: 'Do you agree with that description? Thumbs up / thumbs sideways / thumbs down.' This quick pulse check surfaces disagreement about the shape and seeds discussion for the histogram lesson. Close the lesson by returning to the anchoring phenomenon: 'We started today with a chaotic list of 40 prices. You have now organised it into a frequency distribution table and sketched its shape. Has this changed your answer to our Driving Question — who is right, the family or the neighbour?' Wait 15 seconds of silence. Do not answer — let the question sit.	Cumulative Model Revision (NGSS Science and Engineering Practice 2 — Developing and Using Models): The model page makes the learner's growing understanding tangible and cumulative. Adding a frequency-derived sketch to the model page in Lesson 3 means learners literally see Layer 1 (raw data impression), Layer 2 (previous lesson's initial observations), and Layer 3 (frequency-organised sketch) on the same page — the growth of understanding is visible. The shoulder-partner share of shape descriptions activates peer sense-making and surfaces the vocabulary ('clustered', 'spread', 'peak') that will be formalised in the histogram lesson.	Observable indicator: Each learner's model page shows a dot-plot or bar sketch with correct relative heights (taller columns for higher-frequency prices) and a written shape description that references the Driving Question. The teacher collects 6 exercise books at random at the end of the lesson (chosen from different table groups) and checks: (1) Is the frequency distribution table complete and correctly totalled? (2) Does the sketch reflect the actual frequencies? (3) Does the shape description make a claim about the maize price distribution? These books are returned with a brief written comment before Lesson 4.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. DATA INTEGRITY MOMENT: When the class constructed the shared frequency table on the board, did discrepancies between pairs' tables arise? How were they resolved — by going back to the raw data or by majority vote? If discrepancies were glossed over, plan a 5-minute 'data integrity' opener for Lesson 4 in which you model that in honest statistics, we always return to the source data to resolve disputes — connecting directly to the Driving Question's emphasis on 'telling an honest story.' 2. COGNITIVE CONFLICT: Did learners experience genuine tension when asked in Q2 (Explain phase) whether the table supports the family or the neighbour? If most learners converged quickly on one side without debate, the productive conflict was not activated. In Lesson 4 (mean and median), deliberately choose two different summary statistics that give different impressions of the Wakulima data — this will reactivate the conflict at a deeper level. 3. DQB QUALITY: Read the orange 'wonder' sticky notes carefully. Are learners' new questions pointing toward mean/median/mode and graphical display (the upcoming lessons), or are they veering toward unrelated topics? If the questions are off-track, this suggests the phenomenon connection is weakening — spend the first 3 minutes of Lesson 4 explicitly re-reading the anchoring phenomenon scenario before any new instruction. 4. MODEL BUILDING: Did learners' dot-plot sketches show a recognisable distribution shape? If most sketches were flat (roughly equal heights across all price values), the frequency table may have been constructed with errors. If they show a clear peak, learners are ready for the histogram. Note which learners' shape descriptions used the word 'clustered' or equivalent — these learners are conceptually ready to engage with measures of central tendency as a formalisation of 'where the cluster is.' 5. EQUITY CHECK: Were all pairs equally engaged during the tallying phase, or did one partner dominate? In Lesson 4, restructure pairs and assign explicit roles (Reader / Recorder / Checker) to ensure equitable participation, particularly for learners who were passive talliers today.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners worked with the raw list of 40 daily maize flour prices from Wakulima Market — the same chaotic data from the anchoring phenomenon — and organised it by tallying each price value and recording its frequency in a three-column table (Price | Tally | Frequency). They verified their table summed to 40 entries and compared their pair table to a class-constructed table on the board.
What did I learn?	A frequency distribution table for ungrouped data organises raw data by listing each distinct value alongside a tally and frequency count. The value with the highest frequency is the mode. Organising raw data into a frequency table reveals patterns — such as which prices are common and which are rare — that are completely invisible in the raw list. The sum of the frequency column must always equal the total number of data points (40 in our dataset).
How does this explain the phenomenon?	When we organised the 40 Wakulima Market maize prices into a frequency distribution table, we could identify the modal price — the price that appeared most often. This gave us our first structured evidence to begin answering the driving question: the family who felt prices 'varied wildly' may have visited on low-frequency days at the extremes, while the neighbour who said prices were 'always about the same' may have experienced the high-frequency modal price cluster. The table alone cannot fully settle the argument — we still need measures of central tendency and a graphical display to tell the complete, honest story.

LESSON 4 (80 minutes): Frequency Distribution Tables for Ungrouped Data — Bringing Order to the Maize Price Mystery	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners construct frequency distribution tables for ungrouped data and use them to reveal patterns hidden in the raw Wakulima Market maize price list, advancing the class investigation into who is right — the family or the neighbour.
Knowledge	Learners demonstrate understanding of: (i) the meaning of frequency as the number of times a value occurs in a data set; (ii) the structure of a frequency distribution table (value column, tally column, frequency column); (iii) how a frequency table organises ungrouped data to reveal its distribution; (iv) the distinction between raw data and organised data.
Skills	Learners are able to: (i) collect and record data using tally marks accurately; (ii) construct a complete frequency distribution table for a given ungrouped data set; (iii) read and interpret a frequency distribution table to identify the most common and least common values; (iv) use the frequency table to begin calculating the mean, mode and median of ungrouped data.
Attitudes	Learners appreciate that organising data with integrity — recording every value, not skipping inconvenient prices — is essential for honest and fair decision-making in the community; they value the effort required to move from chaotic raw data to a clear, truthful summary.
Key Inquiry Question	How do we represent data?
Purpose in Storyline	This is Lesson 4 of 8 in the evidence-gathering sequence. Having predicted (Lesson 1), explored raw data and range (Lessons 2–3), learners now encounter their first formal organisational tool — the frequency distribution table. The table immediately makes visible which maize prices occur most often, giving the class their first structured evidence to revisit the family-vs-neighbour argument. This lesson lays the algebraic scaffolding (Σf, Σfx) needed for mean and median calculations in Lessons 5–6.
Safety Notes	No physical hazards. Remind learners to use rulers when drawing table lines to avoid errors that could misrepresent data — integrity in data recording is itself a safety net against bad decisions.

B. LESSON OVERVIEW
Learners begin by re-examining the chaotic raw list of 40 Wakulima Market maize prices from Lesson 1 and predict whether simply re-reading that list can settle the family-vs-neighbour argument. They then observe a teacher-modelled tally process on 10 prices before constructing their own full frequency distribution table for all 40 prices in pairs. A structured class discussion — using the completed tables — surfaces the concepts of frequency, tally, and how the table begins to answer the driving question. Learners add a 'Σfx' column as a bridge to the mean. The lesson closes with a DQB update and a model revision: learners annotate their running data-story model with the frequency table as a new organising layer.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — Predict 📹 **VIDEO:** [Ungrouped Data]	Each learner receives a printed card showing the same chaotic list of 40 Wakulima Market maize prices (in KSh per 2 kg packet)	Display the raw price list on the board (large font or chart paper). Say verbatim: 'You have seen this list before. Today I want you to be	Prediction Activation: surfacing prior intuition before instruction creates cognitive dissonance — learners who cannot extract the	Observable indicator: Do learners' written predictions show awareness that the raw list is hard to read (even if their guess is

to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Line Plots from Frequency Tables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	that was introduced in Lesson 1: e.g. 98, 105, 98, 112, 98, 105, 120, 98, 105, 112, 98, 120, 105, 98, 112, 105, 98, 120, 98, 105, 112, 98, 105, 98, 112, 120, 98, 105, 98, 112, 105, 98, 120, 98, 105, 112, 98, 105, 98, 112. Learners work individually for 3 minutes and answer TWO predict questions in their exercise books: (Q1) Without doing any calculation, just by staring at the raw list, can you tell which price appears most often? Write your best guess. (Q2) Do you think the family (who say prices vary wildly) or the neighbour (who says prices are stable) would be helped by this raw list? Explain in one sentence why or why not.	honest — can you actually read the answer off this list, or is it still a mess?' Allow 3 minutes of silent individual thinking. Circulate and note 3–4 interesting predict responses (one confident correct guess, one wrong guess, one 'I cannot tell'). Do NOT confirm or correct yet. After 3 minutes, cold-call exactly 3 learners: 'Amina, read me your answer to Q1.' — 'Otieno, what did you write for Q2?' — 'Wanjiru, do you agree with Otieno? Why?' Apply 12-second WAIT TIME after each cold-call before accepting the response. Record the three guesses visibly on the board under the heading 'OUR PREDICTIONS — Lesson 4'. Say: 'Hold those predictions. By the end of today, the data itself will tell us who was right.'	answer from the raw list immediately feel the need for a better tool, motivating the frequency table as a solution rather than a procedure to memorise.	wrong)? Listen for language like 'it looks like 98 keeps appearing' or 'I cannot be sure' — both signal productive engagement. Flag any learner who claims the raw list is perfectly clear for follow-up questioning during Phase 2.
Phase 2 — Observe 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Line Plots from Frequency Tables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Teacher live model (10 min): The teacher works through ONLY the first 10 prices from the list on the board, building a tally chart in real time. Learners watch and record the emerging table in their books: columns headed 'Price (KSh)', 'Tally', 'Frequency (f)'. They see tally marks grouped in fives (four vertical strokes, fifth diagonal). After 10 prices, the teacher freezes and asks learners to complete the frequency column themselves before revealing it. PART B — Pair construction (20 min): In assigned pairs, learners extend the same table to cover all 40 prices. Each pair receives a blank three-column table template (ruled, with the four price values 98, 105, 112, 120 already listed in column 1). Learners take turns reading prices aloud while the other marks tallies, then swap roles at price 20. They complete	PART A: Stand at the board. Say: 'Watch exactly what I do — I am going to read this list like a careful Wakulima Market record-keeper.' Read the first price aloud: 'Ninety-eight shillings' — draw one tally mark next to 98. Read the next: 'One hundred and five' — mark next to 105. Continue through price 10, narrating every mark. At price 5 say: 'I have four marks for 98 — watch what happens to the fifth.' Model the diagonal fifth stroke and say: 'That bundle of five makes counting easy later.' After 10 prices, say: 'Now YOU fill in the frequency column before I do.' Give 90 seconds of silent work. Cold-call one pair: 'Kamau and Achieng — what frequency did you get for 98 so far?' Apply 10-second WAIT TIME. Reveal and confirm. PART B: Circulate every 3 minutes. Prompt pairs who are not alternating roles: 'Who is reading	Concrete–Representational connection: the tally mark process makes the abstract concept of 'frequency' physically visible — each stroke represents one real price paid by a real family at Wakulima Market. Alternating reader/marker roles in pairs embeds the data-collection process as a collaborative, error-checking practice (NGSS Science and Engineering Practice 3: Planning and Carrying Out Investigations; Practice 4: Analysing and Interpreting Data).	Observable indicator 1: All frequency values in a correctly completed table should be 16 (for KSh 98), 12 (for KSh 105), 8 (for KSh 112), 4 (for KSh 120) — summing to 40. Circulate and check sums during PART B; any pair whose frequencies do not sum to 40 has a tally error and must recount. Observable indicator 2: In PART C, $\Sigma fx = (16\times98)+(12\times105)+(8\times112)+(4\times120) = 1568+1260+896+480 = 4204$. Pairs who get a different Σfx have an arithmetic error in multiplication — note these pairs for targeted support in Phase 3.

	the frequency column and check that frequencies sum to 40. PART C — Adding the fx column (5 min): Teacher introduces a fourth column headed 'fx' (price × frequency). Learners calculate and fill in fx for each row, then find Σfx (the sum of all fx values). They record Σf = 40 and Σfx at the bottom of the table.	and who is marking right now? Swap.' Spot-check at price 20 by asking: 'What should your four frequencies add up to at this point?' (Answer: 20.) Prompt struggling pairs: 'Point to each tally mark and count — does your tally match your frequency number?' PART C: Write 'fx' on the board. Say: 'This column is our bridge to the mean. fx means: how much money would have been paid in total if every purchase at this price happened f times?' Model the first row: 98 × 16 = 1 568. Ask pairs to complete the remaining three rows. Say: 'When you add up all your fx values, what do you get? Write that number — it is the total shillings paid across all 40 purchases.' Give 3 minutes. Cold-call two pairs for their Σfx value and compare — if they differ, ask: 'How will we find out who is right?' (recount tallies, re-multiply).		
Phase 3 — Explain 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles]() Source: CK-12 🔍 [Search ARES for similar videos]() 📖 **HTML:** [Line Plots from Frequency Tables (Flexbook)]() Source: CK-12 🔍 [Search ARES for similar readings]()	Learners work in groups of four (two pairs combining) to answer a structured Explain worksheet with four questions anchored to the phenomenon: (E1) Which maize price appeared most often at Wakulima Market? What do we call this value in statistics? (E2) A journalist from the Daily Nation wants to write ONE sentence summarising the maize price situation. Using only your frequency table, write that sentence. (E3) The family says prices 'vary wildly'. The neighbour says prices are 'always about the same'. Use your frequency table — give each person ONE piece of evidence from the table that SUPPORTS their view. (E4) Wanjiru says, 'The average price must be KSh 108.50 because that	Distribute the Explain worksheet. Say: 'Questions E1 to E4 — you are not doing arithmetic for its own sake. You are using the table to settle a real argument about real food prices that affect real Nairobi families.' Circulate during group work. For E1, prompt groups that write only the number: 'You have the number — what is the statistical NAME for the most frequent value?' (mode). For E3, push groups who only support one side: 'You found evidence for the family — now play devil's advocate and find evidence for the neighbour.' For E4, write the formula scaffold on the board: Mean = Σfx ÷ Σf = 4204 ÷ 40 = ? Do NOT compute it — let groups calculate. After 10 minutes, facilitate debrief: cold-call one	Claim–Evidence–Reasoning (CER): learners must use specific cells from their frequency table as evidence and connect that evidence to the competing claims of the family and neighbour. This prevents rote answers and forces learners to treat the table as a tool for argumentation, not just a bookkeeping exercise. (NGSS Practice 6: Constructing Explanations and Designing Solutions; Practice 7: Engaging in Argument from Evidence.)	Observable indicator 1: Learners correctly identify mode = KSh 98 and use the word 'mode' (not just 'most common number'). Observable indicator 2: Learners calculate mean = 4204 ÷ 40 = 105.10 and explicitly state that Wanjiru's midrange estimate of 108.50 is NOT the mean because it ignores the unequal frequencies. Observable indicator 3: In E3, listen for whether groups cite BOTH a high-frequency cluster (supporting the neighbour) AND the presence of KSh 120 prices (supporting the family) — groups that find only one side need a follow-up prompt in Phase 4.

	is halfway between 98 and 120.' Use Σfx and Σf from your table to calculate the actual mean. Is Wanjiru right? Why or why not? Groups discuss for 10 minutes, then the teacher facilitates a 10-minute whole-class debrief.	group for E1 ('Njoroge — what is the mode and how does the table prove it?'), one group for E3 ('Adhiambo — read us the two pieces of evidence you found'), one group for E4 ('Mwangi — what mean did your group calculate? Now is Wanjiru right?'). Apply 12-second WAIT TIME after each question. Close the Explain phase by writing on the board: 'FREQUENCY TABLE INSIGHT: The most common price was KSh 98 — it appeared 16 out of 40 times (40%). This table is now our first organised evidence in the Maize Price Mystery.' Ask: 'Does this one fact alone settle the argument? Vote — thumbs up if yes, thumbs sideways if maybe, thumbs down if no.' Count and record the vote visibly.		
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Line Plots from Frequency Tables (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner individually writes ONE sticky note (or a designated DQB section in their book if physical sticky notes are unavailable) responding to the prompt: 'What does today's frequency table tell us about the Maize Price Mystery — and what does it still NOT tell us?' Learners post or record their sticky notes under two headings on the class DQB: 'NOW WE KNOW…' and 'WE STILL WONDER…'. The class reads 3–4 selected notes aloud and votes (show of hands) on which unanswered question is most important to investigate next. The teacher also returns to the three Lesson 1 predictions posted on the DQB and asks: 'Can we now confirm or revise any of these older predictions?'	Point to the DQB. Say: 'The DQB is our living record of how our thinking is growing. In Lesson 1 you predicted whether prices varied or were stable. In Lesson 4 you have your first hard evidence — a frequency table. Let's update our board honestly.' Give 3 minutes of silent individual writing. Circulate and read notes — select 2 'NOW WE KNOW' notes that correctly use the word 'mode' or 'frequency' and 2 'WE STILL WONDER' notes that point toward mean/median/histogram (without necessarily using those words). Read each selected note aloud (with the author's permission) and ask the class: 'Do you agree this belongs under NOW WE KNOW / WE STILL WONDER?' After posting, point to the Lesson 1 predictions still on the board. Say: 'Amina predicted in Lesson 1 that the price was always the same.	DQB as Epistemic Tracking: the dual-column structure ('Now We Know' / 'We Still Wonder') makes metacognitive progress visible and collective. Revisiting Lesson 1 predictions models scientific practice — scientists update claims when new evidence arrives. (NGSS Practice 8: Obtaining, Evaluating and Communicating Information.)	Observable indicator: 'NOW WE KNOW' entries should reference concrete table findings (e.g. 'KSh 98 appears most often', 'mode = 98', 'mean ≈ 105'). 'WE STILL WONDER' entries should point toward unresolved aspects (e.g. 'why do prices jump to 120 sometimes?', 'does a bar chart show this better?', 'what is the middle price?'). Any learner whose DQB entry is still purely about the raw list (not the frequency table) has not yet internalised the lesson's central organising insight and needs individual follow-up.

		Can we now confirm, revise or reject that prediction?' Cold-call Amina if she is present. Apply 12-second WAIT TIME. Close by saying: 'Our DQB now shows that we have one organisational tool — the frequency table. Our story is getting clearer, but it is not finished. What might we need next to see the SHAPE of this data more clearly?' Accept 2–3 responses without confirming — plant the idea of graphical display for Lesson 5.		
Phase 5 — Model Building 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Line Plots from Frequency Tables (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners open the 'Maize Price Story Model' they have been building since Lesson 1 (a running annotated diagram/table in a dedicated section of their exercise books). Today's model addition task: (M1) Paste or carefully copy the completed frequency distribution table (with tally, f, and fx columns) into the model. (M2) Using a ruler, draw a simple dot-plot or vertical line plot of the frequency data directly below the table — mark the four price values on a horizontal axis, and for each value draw a vertical stack of dots equal to its frequency (e.g. 16 dots above KSh 98). (M3) Annotate the model with two labelled arrows: one pointing to the tallest stack labelled 'MODE = KSh 98 (most common price)' and one pointing to the calculated mean labelled 'MEAN = KSh 105.10'. (M4) Write one 'Model Update Sentence': 'After Lesson 4, our frequency table shows that maize prices at Wakulima Market are NOT spread randomly — most purchases (40%) were at KSh 98, which supports the neighbour's view, but the presence of KSh 120 prices supports the family's view that	Say: 'Your model is the story you are building — each lesson adds a new chapter. Today's chapter is called ORGANISATION.' Display a sample model page under a visualiser or on the board (hand-drawn is fine). Walk through M1–M4: 'M1 — copy the table neatly. M2 — the dot-plot is a preview of something more powerful we will build in Lesson 6. M3 — label the mode and the mean so any reader of your model instantly sees the two key summary numbers. M4 — write the update sentence in YOUR own words — do not copy mine exactly.' Give 8 minutes. Circulate and check: (a) that the table is complete and accurate; (b) that the dot-plot has the correct number of dots per price; (c) that the mode and mean labels are correct. For learners who finish early, ask: 'If you had to add a third label — one that the neighbour would point to and one the family would point to — where would you put them?' Close the lesson by saying: 'In Lesson 5, we will ask: if the mean is KSh 105.10 but the mode is KSh 98, which number gives the family the FAIREST answer? That question will	Cumulative Model Revision: requiring learners to physically integrate today's frequency table into a model built across eight lessons embeds the understanding that statistics is a progressive story, not a series of isolated techniques. The dot-plot preview creates a visual bridge from tabular to graphical representation, reducing cognitive load when histograms are introduced in Lesson 6. (NGSS Practice 2: Developing and Using Models.)	Observable indicator 1: The completed model page contains all four elements (M1–M4) with the frequency table summing to f = 40, the dot-plot having 16+12+8+4 = 40 dots, and mode/mean correctly labelled. Observable indicator 2: The Model Update Sentence (M4) correctly attributes the mode as evidence for the neighbour AND acknowledges the high-price tail as evidence for the family — demonstrating the lesson's core conceptual tension. Collect and scan 6–8 model books at the end of class (select a range of abilities) to inform the opening of Lesson 5.

	prices can spike.'	introduce us to the median.' Leave this on the board as a bridge to Lesson 5.		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) During Phase 2, did all pairs achieve a frequency table that summed to exactly 40? If more than 25% had tally errors, consider beginning Lesson 5 with a 5-minute 'tally audit' before introducing median. (2) In Phase 3 (E3), could learners find evidence for BOTH the family AND the neighbour from the same table? If most groups only argued one side, the concept of multiple valid interpretations of the same data set needs re-emphasis — it is the heart of the driving question. (3) Did the mean calculation ($\Sigma fx \div \Sigma f = 105.10$) land as meaningful or as mere arithmetic? If learners could not explain why Wanjiru's midrange (108.50) was wrong, they have not yet connected the frequency weights to the mean formula — plan a short re-engagement in Lesson 5. (4) Review the DQB 'WE STILL WONDER' entries: do they naturally generate questions about median and graphical display? If learners are already asking 'what does a bar chart of this look like?' or 'which price is in the middle?', you can accelerate the Lesson 5 opening. (5) Core competency check — Integrity: did you observe any pair that appeared to fabricate or round their tally counts to make the sum reach 40? Address this privately and connect it to the lesson's values: honest data is the foundation of decisions that affect families buying food at Wakulima Market.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed the same 40 raw Wakulima Market maize prices from Lesson 1 being transformed — through a systematic tally process — into a frequency distribution table with four price values (KSh 98, 105, 112, 120), their tally marks, frequencies (16, 12, 8, 4), and a calculated fx column summing to $\Sigma fx = 4204$.
What did I learn?	Learners learned that: (i) frequency means the number of times a value occurs in a data set; (ii) a frequency distribution table organises raw data into a structure that immediately reveals which values are common and which are rare; (iii) the mode is the value with the highest frequency; (iv) the mean of ungrouped data can be calculated as $\Sigma fx \div \Sigma f$, and this weighted mean (KSh 105.10) differs from a simple midrange estimate because it accounts for how often each price actually occurred.
How does this explain the phenomenon?	Learners explained how the frequency table provides the first structured evidence in the Maize Price Mystery: the mode (KSh 98, occurring 40% of the time) supports the neighbour's view that prices are stable, while the presence of KSh 120 prices (occurring 4 times) supports the family's experience of occasional expensive days — showing that both parties were telling the truth from different statistical perspectives. Learners also explained why organising data with integrity (recording every value accurately) is essential for fair community decision-making.

LESSON 5 (80 minutes): Histograms: Making the Maize Price Distribution Visible	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to represent grouped data using histograms and to interpret what the shape of a histogram reveals about a dataset — directly applied to the Wakulima Market maize flour price data.
Knowledge	Learners will know: (i) that a histogram uses bars whose AREA (not just height) represents frequency; (ii) how to calculate frequency density = frequency ÷ class width; (iii) that equal class-width histograms have bars of equal width while unequal class-width histograms require frequency density on the y-axis; (iv) how to read and interpret the shape (skew, spread, modal class) of a histogram to draw conclusions about a real dataset.
Skills	Learners will be able to: (i) accurately draw a histogram for both equal and unequal class-width grouped data from the Wakulima maize price dataset; (ii) calculate frequency density for each class; (iii) correctly label axes, scale, and title a histogram; (iv) interpret the shape of the histogram to make evidence-based claims about price variability vs. price stability.
Attitudes	Learners will appreciate that a well-constructed histogram makes a complex dataset immediately understandable — turning a chaotic list of 40 prices into a visual story — and will value integrity in accurate data representation as a tool for honest community decision-making.
Key Inquiry Question	How do we represent data? (KICD KIQ 2) | How do we use statistics in day-to-day life? (KICD KIQ 3)
Purpose in Storyline	In Lessons 1–4 learners collected, organised, and summarised the 40 Wakulima maize prices using frequency distribution tables (ungrouped and grouped) and calculated mean, mode, and median. They discovered that different measures of centre gave subtly different answers — leaving the family-vs-neighbour argument unresolved. In Lesson 5 learners now DRAW a histogram of the grouped data. The visual shape of the histogram produces the 'aha' moment: the distribution is right-skewed, with most prices clustered in a low band and a thin tail of occasional high prices. This explains BOTH the neighbour's experience (most days prices are similar) AND the family's experience (rare high-price days feel shocking). The histogram becomes the most powerful piece of evidence gathered so far.
Safety Notes	No hazardous materials. Ensure rulers have no sharp edges. If using printed graph paper, handle scissors safely when trimming. Remind learners to keep workspaces tidy to avoid trips during group movement.

B. LESSON OVERVIEW

Lesson 5 of 8 in the evidence-gathering sequence for the Wakulima Market maize price driving question. Learners move from numerical summaries (Lessons 1–4) to graphical representation. Using the grouped frequency distribution table they constructed in Lesson 3, learners first predict what a bar-chart-style graph of the prices will look like, then observe a teacher-modelled histogram, then construct their own histograms (equal class-width version first, then an unequal class-width extension), and finally interpret the resulting shape. The lesson culminates in updating the Driving Question Board with the new visual evidence and revising the cumulative data model. KICD core competencies addressed: Critical Thinking and Problem Solving (interpreting histogram shape), Communication and Collaboration (group histogram construction and gallery walk), Digital Literacy (optional GeoGebra/spreadsheet extension). Values: Integrity (accurate plotting), Unity (group work). PCIs: Life Skills — Decision Making; Parental Engagement (learners are encouraged to share histograms with family members who shop at Wakulima or local markets).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict (10 minutes) 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Baby Due Date Histogram (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the grouped frequency distribution table of the 40 Wakulima maize flour prices (built in Lesson 3) projected on the board. Without any instruction on histograms, each learner individually sketches in their exercise book what they think a graph of this table would look like — 'If you were going to draw a picture of this table so that someone who has never seen the numbers could immediately understand the prices, what would it look like?' Learners annotate their sketch with a one-sentence prediction: 'I think the graph will show that prices are … because …'	Display the grouped FDT (class intervals: KSh 90–99, 100–109, 110–119, 120–129, 130–139, 140–149 with their frequencies). Say exactly: 'Before I teach you anything about histograms, I want you to draw what YOU think the best picture of this table looks like. You have 4 minutes — work alone and in silence.' [START TIMER]. After 4 minutes: 'Now write one sentence predicting what the shape of your graph tells us about the maize prices.' After 5 minutes total, cold-call 4 learners (select gender-balanced, mix of confidence levels) using lolly sticks. Ask each: 'What did you draw and what does the shape say about the prices?' Apply 12-second WAIT TIME after each response before probing. Record key prediction words on the board (e.g. 'tall bar in the middle', 'leans to the left', 'all bars same height'). Do NOT confirm or correct yet — say: 'Hold these predictions. By the end of the lesson, you will judge which were closest to the truth.'	PREDICT-BEFORE-INSTRUCTION (PBI): Activates prior knowledge of bar charts from Junior School and creates cognitive readiness. By committing a prediction to paper, learners have a personal stake in the upcoming observation, increasing engagement and retention of the 'aha' moment when the correct histogram is revealed.	Circulate during the 4-minute sketch phase. Observable indicators of readiness: (i) learner places class intervals on x-axis and frequency on y-axis — shows transfer from bar chart knowledge; (ii) learner draws bars of equal width — shows intuitive understanding that will be refined; (iii) learner's predicted shape (whether correct or not) is annotated with reasoning tied to the price data. Flag learners who draw a pie chart or line graph — these learners will need additional scaffolding in Phase 2.
Phase 2 — Observe (20 minutes) 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Baby Due Date Histogram (PLIX) Source: CK-12	The teacher models constructing a histogram step-by-step on the board using the maize price data (equal class-width, width = 10 KSh). Learners follow along in their exercise books, constructing their own histogram simultaneously. A second dataset (unequal class widths — a 'messy' version of the maize data where a market vendor merged two classes) is then displayed. Learners observe why the simple bar-height method gives a MISLEADING graph for unequal	Step 1 (8 min) — Equal class-width histogram: Draw x-axis labelled 'Price of 2 kg maize flour (KSh)' and y-axis labelled 'Frequency (Number of days)'. Plot class boundaries (89.5, 99.5, 109.5 … 149.5) — pause and ask: 'Why do I use 89.5 and not 90 as my first boundary? Discuss with the person next to you for 30 seconds.' Cold-call 2 pairs. Draw bars with NO gaps, emphasising: 'Histograms have no gaps because prices are continuous data — a price of KSh 109.7 is	WORKED EXAMPLE WITH PARALLEL CONSTRUCTION: Teacher models each step while learners simultaneously replicate it — a 'gradual release' approach. The deliberate introduction of the unequal class-width ERROR before the correction is a productive cognitive conflict strategy: learners feel the need for frequency density rather than being told to use it. Connecting each feature (tallest bar, skew, tail) back to the Wakulima phenomenon ensures the	Circulate during parallel construction. Check: (i) Are bars touching? (no gaps = continuous data understood); (ii) Is the x-axis labelled with class boundaries, not midpoints? (iii) Are y-axis values evenly spaced? (iv) For the unequal class-width section — can the learner correctly compute frequency density for the merged class 90–109 (frequency ÷ 20)? Target question for struggling learners: 'Show me how you worked out the height of this bar.' Target extension question: 'What

🔍 Search ARES for similar readings	widths and see frequency density introduced as the correction. Learners complete a structured observation table: 'What I see in the histogram | What it means for the maize prices | Question it raises for me.'	possible.' Complete all 6 bars. Step 2 (6 min) — Reveal the unequal class-width problem: Show a second table where a vendor grouped data as: 90–109 (merged), 110–119, 120–129, 130–149 (merged). Ask: 'If I draw bars using frequency as height, which bar looks biggest?' Wait 10 seconds. Cold-call 1 learner. Then say: 'But is the merged bar really that much more common, or is it just wider? This is where the graph would LIE to us.' Introduce frequency density: frequency density = frequency ÷ class width. Work through all four merged classes live. Redraw corrected histogram with y-axis labelled 'Frequency Density'. Step 3 (6 min) — Observation table: Learners complete the three-column observation table individually, connecting each histogram feature to the Wakulima phenomenon. Say: 'Look at the shape. Where is the tallest bar? What does that tell the family and the neighbour?'	mathematics is always in service of real sense-making.	would happen to the histogram if we merged ALL classes into one? What information would we lose about the maize prices?'
Phase 3 — Explain (20 minutes) 📹 **VIDEO:** Experimental probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Baby Due Date Histogram (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In groups of 4, learners receive a fresh dataset: the 40 maize prices re-sorted with a DIFFERENT set of class intervals (width = 15 KSh: 85–99, 100–114, 115–129, 130–144, 145–159) to create unequal-width classes relative to their Lesson 3 table. Each group constructs a histogram on A3 graph paper using frequency density on the y-axis. Groups then write a 3-sentence 'Verdict Statement' on a sticky note: 'Based on our histogram, the neighbour is [right/wrong/partly right] because … The family is [right/wrong/partly right] because … Therefore, the honest answer to who is right is …' Groups post their	Distribute A3 graph paper, rulers, and coloured pencils. Say: 'Your group has 12 minutes to construct a histogram using frequency density. Every member of the group must be able to explain ANY bar — I will cold-call individuals, not groups.' [CIRCULATE — spend no more than 90 seconds per group]. Prompt struggling groups: 'Before you draw anything, tell me: what goes on the x-axis? What goes on the y-axis? What is the class width of your first bar?' After 12 minutes: 'Stop drawing. Write your 3-sentence Verdict Statement in the next 3 minutes.' Gallery Walk (5 min): Each group leaves one 'host' at their poster;	GALLERY WALK WITH CRITIQUE DOTS: Making thinking visible on the wall externalises reasoning and allows peer assessment without the social pressure of direct debate. The 'host' role ensures every learner can articulate the mathematics, not just the fastest group member. The final whole-class discussion about 'different verdicts from the same data' directly addresses the core phenomenon tension and foreshadows the upcoming lesson on interpretation and bias in statistics.	During group work, use a clipboard checklist. For each group record: (i) frequency density correctly calculated for all bars (Y/N); (ii) y-axis labelled 'Frequency Density' (Y/N); (iii) bars drawn to correct height with no gaps (Y/N); (iv) Verdict Statement references specific histogram features (Y/N). During Gallery Walk, note which learners can verbally justify the shape of the histogram to rotating visitors. Red-dot disagreements that generate discussion are high-value formative data — identify the specific misconception (e.g. confusing frequency with frequency density, or misreading bar height as area).

	histograms and verdict statements on the classroom walls for a Gallery Walk.	remaining members rotate. Hosts explain their histogram and verdict. Rotating members place a green dot sticker if they agree with the verdict or a red dot if they disagree. After the walk: ask 2 groups with contrasting verdicts to defend their reasoning. Pose to the class: 'Two groups looked at the same data and reached different verdicts. Is that a problem? What does it tell us about using statistics to settle arguments?' Apply 15-second WAIT TIME. Cold-call 3 learners.		
Phase 4 — Driving Question Board Update (10 minutes) 📹 **VIDEO:** [Experimental probability]() Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Baby Due Date Histogram (PLIX)]() Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board displayed at the front. Each learner writes on a coloured card: one CLAIM they can now make about the maize prices supported by the histogram evidence, and one NEW QUESTION the histogram has raised. Cards are sorted live into two columns: 'What We Now Know' and 'What We Still Need to Find Out.' Learners identify which prior questions from the DQB (posted in Lessons 1–4) have been answered by the histogram and physically move those cards to a 'Resolved' pocket.	Distribute pre-cut cards (yellow for claims, blue for new questions). Say: 'You have 3 minutes. On the yellow card write: I can now claim that [specific statement about maize prices] because the histogram shows [specific feature — e.g. the tallest bar, the tail, the skew]. On the blue card write a NEW question this lesson has made you wonder about.' Collect cards. Read 5–6 aloud without attribution and sort them to the DQB columns live. Ask: 'Which question from our board in Lesson 1 does today's histogram finally help us answer?' Wait 12 seconds. Cold-call 2 learners. Move resolved cards to the 'Resolved' pocket with a tick. Point to remaining unanswered questions: 'These are what Lessons 6, 7 and 8 will help us resolve. Notice that our histogram has answered the SHAPE of the price distribution — but has it told us everything? What about comparing two different markets, or two different months?' This plants the seed for future work on interpretation.	DRIVING QUESTION BOARD — CLAIM-EVIDENCE-QUESTION CYCLE: The structured card routine (yellow = claim tied to evidence, blue = new question) enforces the Science and Engineering Practice of Constructing Explanations with Evidence. Publicly moving 'resolved' cards gives learners a concrete sense of growing understanding and maintains the narrative momentum of the storyline.	Yellow cards are collected and quickly scanned. Look for: (i) Does the claim cite a SPECIFIC histogram feature (e.g. 'the bar for KSh 100–109 is tallest') rather than a vague statement ('prices are low')? (ii) Does the claim connect back to the phenomenon (family vs. neighbour)? (iii) Is the claim proportionate — does it over-claim certainty (e.g. 'this proves prices never change') or is it appropriately hedged ('most of the time prices fall between …')? Learners who over-claim will be targeted in Lesson 6 for work on the limitations of graphical representation.
Phase 5 —	Learners retrieve the cumulative	Distribute learners' individual	CUMULATIVE MODEL	Collect Model Update Statements

Model Building (20 minutes) 📹 **VIDEO:** [Experimental probability] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Baby Due Date Histogram (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	data model they have been building since Lesson 1 (a large annotated diagram showing how evidence about maize prices is building up). They add a new layer: a miniature sketch of their histogram glued or drawn onto the model, with three annotation arrows pointing to: (1) the modal class and what it means for the 'average' shopping day, (2) the right-skewed tail and what it means for the family's 'shock' high-price experiences, and (3) the spread (range of bars) and how it relates to the neighbour's claim. Learners then write a 'Model Update Statement': 'Before today I thought the data showed … After drawing and interpreting the histogram, my model now shows … The piece of evidence that most changed my thinking was …'	model folders (stored from previous lessons). Say: 'You have 15 minutes to update your model. The histogram is your new evidence layer. Your three annotation arrows must each use a specific number from the histogram — not just general words.' Circulate and prompt: 'What number from the frequency density axis will you write on arrow 2? Can you connect the skew directly to something the FAMILY said about their shopping experience?' After 13 minutes: 'Stop and share your Model Update Statement with the person next to you. Your partner should check: did you use a specific histogram feature? Did you connect it to the Wakulima family or neighbour?' After 2 minutes of pair share: Cold-call 3 learners to read their Model Update Statement aloud. Offer specific praise: 'I like that you used the word skewed — that is a precise statistical word. Can you explain in one sentence what right-skewed means to someone who has never heard that word?' Close with: 'Next lesson we will explore frequency polygons — another way to visualise the same data. Ask yourself tonight: if you could draw a LINE version of this histogram, what new things might it show?'	REVISION: The annotated model is a living document that makes learning progression concrete and visible. Requiring specific numbers from the histogram (not just vague descriptions) forces precision and guards against superficial annotation. The 'Before/After/Most Changed' structure of the Model Update Statement is a metacognitive routine that helps learners articulate conceptual change — a key NGSS Science and Engineering Practice (Obtaining, Evaluating, and Communicating Information). Pair-checking the statement embeds peer accountability.	at the end of class (or photograph models). Use a 3-level rubric: LEVEL 3 (Meets/Exceeds) — statement cites specific histogram features with correct statistical vocabulary (modal class, frequency density, skew, tail), connects features to both the family AND neighbour experience, and identifies a genuine conceptual shift. LEVEL 2 (Approaching) — statement references the histogram but uses general language ('the tall bar', 'the prices go up'), connects to only one party in the phenomenon. LEVEL 1 (Below) — statement is a procedural description ('I drew a histogram') with no interpretive connection to the phenomenon. Learners at Level 1 are flagged for small-group follow-up at the start of Lesson 6.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions before planning Lesson 6:

1. HISTOGRAM CONSTRUCTION ACCURACY — What percentage of groups correctly calculated frequency density for all bars? If fewer than 70% succeeded, begin Lesson 6 with a 5-minute worked example using a fresh Kenyan context (e.g. rainfall data from Kericho tea farms) before moving to frequency polygons.

2. THE 'AHA' MOMENT — Did the Gallery Walk produce genuine discussion about histogram shape and the family-vs-neighbour phenomenon, or did groups merely describe bar heights without interpretation? If discussion was shallow, consider adding a 'Devil's Advocate' role card in future Gallery Walks — one group member must

argue the OPPOSITE of the group's verdict to force deeper reasoning.

3. EQUITY IN PARTICIPATION — Did female learners take the 'host' role in Gallery Walk as often as male learners? Did cold-calling reach learners from all ability bands? Review your lolly-stick log and adjust cold-call patterns in Lesson 6 if any subgroup was underrepresented.

4. DQB QUALITY — Were learners' yellow claim-cards specific and evidence-linked? If many were vague, introduce a sentence frame in Lesson 6: 'The histogram shows [specific feature], which means [interpretation], which suggests that [conclusion about the phenomenon].'

5. MODEL DEPTH — Did learners' annotation arrows use precise vocabulary (frequency density, modal class, right skew)? Note any vocabulary gaps and build a 'Statistics Word Wall' card for those terms to be displayed for the remainder of the unit.

6. UNEQUAL CLASS-WIDTH UNDERSTANDING — The unequal class-width extension is the most cognitively demanding element. Were all learners able to articulate WHY frequency density must be used? If not, plan a short misconception-address segment at the top of Lesson 6 using a deliberately misleading equal-height histogram and asking 'What is wrong with this graph?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a teacher-modelled histogram of the 40 Wakulima maize flour prices (equal class-width), then observed a second demonstration showing how a histogram with unequal class widths MISLEADS when bar height = frequency rather than frequency density. Learners saw that the shape of the resulting histogram is right-skewed — with the modal class in the KSh 100–109 range and a thin tail extending toward KSh 140–149.
What did I learn?	Learners learned that: (i) a histogram uses bars whose AREA represents frequency, so frequency density (frequency ÷ class width) must be used on the y-axis when class widths are unequal; (ii) histograms have no gaps because the data (price) is continuous; (iii) the shape of a histogram (skew, spread, modal class) carries meaning — a right-skewed distribution explains why the neighbour experiences 'stable' prices on most days while the family occasionally encounters shockingly high prices on rare days; (iv) the same data can look different depending on how class intervals are chosen, raising questions about honest data representation.
How does this explain the phenomenon?	Learners explained, in group Verdict Statements and individual Model Update Statements, that BOTH the Wakulima family and their neighbour are telling the truth: the histogram shows that most of the 40 days had prices clustered in the lower classes (supporting the neighbour's experience of stability), but the right-skewed tail confirms that occasional high-price days do exist and feel dramatic when experienced (supporting the family's experience of variability). The histogram provides stronger evidence than the mean, mode, or median alone because it shows the FULL SHAPE of the distribution, not just a single summary number.

LESSON 6 (80 minutes): Grouping the Maize Price Data: Grouped Frequency Tables and the Modal Class	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners construct grouped frequency distribution tables, identify the modal class, and compare grouped versus ungrouped representations — developing the critical understanding that grouping gains clarity but loses detail.
Knowledge	Learners know that: (a) data can be organised into class intervals to create a grouped frequency distribution table; (b) the modal class is the class interval with the highest frequency; (c) grouping data involves a trade-off between clarity of pattern and loss of individual data values.
Skills	Learners are able to: (a) choose appropriate class width and class intervals for a given data set; (b) draw a grouped frequency distribution table for ungrouped data; (c) identify the modal class from a grouped frequency distribution table; (d) compare grouped and ungrouped representations of the same data set and articulate what each reveals or conceals.
Attitudes	Learners appreciate that the choice of class width is a mathematical decision with real consequences — different groupings can tell different stories about the same data, reinforcing integrity in data representation.
Key Inquiry Question	How do we represent data?
Purpose in Storyline	In Lessons 1–5 learners collected the 40 Wakulima Market maize prices, organised them into an ungrouped frequency table, and computed mean, mode, and median. They saw that the raw list looked chaotic. Now, in Lesson 6, they group the same data into class intervals (Ksh 40–49, 50–59, …) to make the distribution visible at a glance. They discover that the modal class does not always match the ungrouped mode, deepening their understanding of what summary statistics actually capture — and moving the class closer to answering the driving question fairly.
Safety Notes	No physical hazards. Remind learners to use pencils for table construction so errors can be corrected neatly. Ensure no learner is ridiculed for choosing a class width that produces an unhelpful table — frame all choices as productive mathematical experiments.

B. LESSON OVERVIEW
Lesson 6 of 8 in the evidence-gathering sequence. Learners return to the full 40-value Wakulima Market maize price dataset and the Kisumu monthly rainfall dataset (supporting phenomenon, introduced in Lesson 4). Working in small groups, they first predict what will happen to information when data is grouped, then physically sort tally cards into class-interval bins to construct grouped frequency tables by hand. They debate class-width choices (Ksh 5 vs Ksh 10 vs Ksh 20 intervals) and observe how the same data can look 'stable' or 'variable' depending on grouping — directly connecting to the neighbour-vs-family dispute in the anchoring phenomenon. The modal class is formally introduced and contrasted with the ungrouped mode. Learners then apply the same process independently to the Kisumu rainfall data and compare the two representations. The lesson closes with a Driving Question Board update and a revision of learners' cumulative data stories.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 —	Learners are shown the 40 Wakulima Market maize prices	Display the 40 prices. Say verbatim: 'Before we do any	Prediction-first activation (NGSS SEP 2 — Developing and Using	Observable indicator: Learners' written predictions reference

Predict 📹 **VIDEO:** Subtracting with Grouped Pictures Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Grouped Data to Find the Mean (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	written in random order on the board (same chaotic list from Lesson 1). The teacher poses: 'We have already found the mean, median, and mode of these prices. Now imagine you sort these 40 prices into groups — say, all prices between Ksh 40 and 49 in one group, Ksh 50 and 59 in another, and so on. In your exercise book, predict: (a) How many groups do you think you will end up with? (b) Which group do you think will be the largest? (c) Will grouping make it easier or harder to see whether the family or the neighbour is right?' Learners write individual predictions (3 minutes), then share with a shoulder partner (2 minutes). Two or three pairs share whole-class. Teacher records predictions on a corner of the board labelled 'Our Predictions — we will test these.'	mathematics, I want your honest gut feeling. Look at these numbers — all 40 of them — and predict what grouping will do to them.' Allow 10 seconds of silent thinking (WAIT TIME) before learners write. After partner sharing, cold-call exactly 3 pairs using names from the register: one who predicts grouping helps the family's case, one who predicts it helps the neighbour's case, one who is unsure. Record all three views without evaluating. Say: 'None of these predictions is wrong yet — we are scientists today. Let the data decide.'	Models): By committing a prediction to paper before any instruction, learners activate prior knowledge about data distribution and create personal cognitive stakes in the outcome. The three competing predictions on the board create productive tension that the Observe phase will resolve.	specific class intervals or frequency estimates rather than vague answers ('more than 4 groups', 'the Ksh 50–59 group will be biggest'). If most predictions are vague or blank, the teacher pauses and reteaches the idea of class intervals using a quick analogy: 'Think of sorting maize sacks by weight — 0–10 kg, 11–20 kg. What would you call each pile?' Collect 4–5 books at random to check written predictions before moving on.
Phase 2 — Observe 📹 **VIDEO:** Subtracting with Grouped Pictures Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Grouped Data to Find the Mean (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group of 4 receives a set of 40 small paper cards, each printed with one of the 40 maize prices (pre-cut by teacher or printed on a strip for learners to cut). Groups also receive three printed 'sorting mats': Mat A uses class width = Ksh 5 (intervals: 40–44, 45–49, 50–54, …), Mat B uses class width = Ksh 10 (40–49, 50–59, …), Mat C uses class width = Ksh 20 (40–59, 60–79, …). Each group is assigned ONE mat. They physically place each price card into the correct class interval, count the cards in each interval, and construct a grouped frequency distribution table in their exercise books, including a tally column. Groups then do a 'gallery rotation' (4 minutes): they view the tables of groups using the other two class widths and note differences. As a	Distribute materials and say: 'Your job is to be data detectives. Sort every price card into the correct interval on your mat — no card left behind. Then build your grouped frequency table.' Circulate every 3 minutes. At the 10-minute mark, stop the class and ask: 'Group using Mat B, what did you find is the most frequent class interval?' (WAIT TIME 12 seconds.) Cold-call the reporter for one Mat B group. Then ask a Mat A group: 'Did you get the same modal class as Mat B?' Cold-call one Mat A reporter. Then ask the Mat C group: 'What happened to the detail when you used Ksh 20 intervals?' (WAIT TIME 10 seconds.) Record the three different modal classes (or confirmations) on the board side-by-side. During gallery rotation,	Concrete Manipulation → Tabular Abstraction (NGSS SEP 3 — Planning and Carrying Out Investigations; SEP 4 — Analysing and Interpreting Data): Physical sorting of cards makes the abstract idea of class intervals tangible. Learners 'feel' the data moving into bins before they write numbers in a table. The three-mat comparison is a controlled investigation into class-width choice, directly observable and discussable. The Kisumu transfer task checks that the skill is not bound to one context.	Observable indicator 1: Each group's grouped frequency table has a tally column that sums correctly to 40 total prices. Teacher checks two groups' tallies before gallery rotation begins. Observable indicator 2: During gallery rotation, learners can verbally articulate at least one difference between the three tables (e.g., 'Mat C hides whether prices were Ksh 40 or Ksh 58 — they look the same'). Observable indicator 3: Individual Kisumu rainfall tables show correct class boundaries and a frequency column that sums to 24. Learners who cannot start the Kisumu task independently are paired with a peer who completed their maize table accurately.

	supporting observation, each learner also receives a printed strip of 24 monthly Kisumu rainfall figures (mm) and individually constructs a grouped frequency table using class width = 20 mm (0–19, 20–39, 40–59, …) in their exercise books.	prompt with: 'Ask yourself — if I were the family trying to prove prices vary, which table would I show? If I were the neighbour trying to prove stability, which table would I show?' For the Kisumu rainfall task, say: 'Now apply exactly the same skill to a different Kenyan dataset — monthly rainfall recorded at Kisumu Meteorological Station.'		
Phase 3 — Explain 📹 **VIDEO:** [Subtracting with Grouped Pictures] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Grouped Data to Find the Mean (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Whole-class discussion anchored by the three grouped frequency tables on the board (one from each mat type). The teacher formally introduces the term 'modal class' and writes the definition: 'The modal class is the class interval with the highest frequency in a grouped frequency distribution.' Learners identify the modal class for each of the three maize price tables and record all three in their exercise books. The teacher then projects (or draws) the ungrouped mode found in Lesson 3 beside the three modal classes, and asks: 'Are they all the same? What does this tell us?' Learners write a 3-sentence 'Data Story' in their exercise books: Sentence 1 — state the modal class for Ksh 10 intervals. Sentence 2 — compare it to the ungrouped mode. Sentence 3 — explain what this means for the family-vs-neighbour dispute. Pairs share their data stories. Selected stories are read aloud and the class agrees on a 'best evidence' sentence to add to the class Data Wall. Learners also identify the modal class for their Kisumu rainfall grouped table and annotate it with one sentence about what it reveals about Kisumu's rainfall pattern.	Write on the board: 'Modal Class = the class interval with the highest frequency.' Say: 'This is NOT the same as the mode of ungrouped data — it is a range, not a single value. Write both the definition and an example in your own words.' Give 3 minutes of silent writing. Then ask: 'Raise your hand if your maize price modal class using Ksh 10 intervals is Ksh 50–59.' Count hands. 'Raise your hand if it is different.' Investigate any discrepancy by asking one learner to walk through their tally aloud — this is a live error-correction moment, not punishment. Cold-call 3 learners to read their Data Story Sentence 3 aloud. After each, ask the class: 'Does this sentence answer our driving question? Does it help settle the family-neighbour dispute? Thumbs up / sideways / down.' Record the consensus sentence on the Data Wall. For the Kisumu modal class, cold-call 2 learners and ask: 'What does your modal class tell a Kisumu farmer about when to plant?'	Concept Definition + Contrast (NGSS SEP 6 — Constructing Explanations): Learners construct the meaning of 'modal class' by comparing it explicitly to the ungrouped mode they already know. The Data Story writing task forces integration of new vocabulary into a real narrative about the phenomenon, making the concept functional rather than inert. Connecting the Kisumu rainfall modal class to farming decisions embeds statistical reasoning in authentic Kenyan life.	Observable indicator 1: Learners can correctly identify the modal class from a grouped frequency table without prompting (check during silent writing). Observable indicator 2: Learners' Data Story Sentence 2 correctly states whether the modal class interval contains the ungrouped mode value. Observable indicator 3: At least 70% of thumbs-up responses to Data Story Sentence 3 indicate class consensus that the sentence is relevant to the driving question. Learners who write sentences disconnected from the phenomenon are redirected with: 'Does your sentence mention the family, the neighbour, or the maize price? If not, revise it.'

Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** Subtracting with Grouped Pictures Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Grouped Data to Find the Mean (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (or writes in a designated DQB section of their exercise book for offline settings). On the FIRST note (green / left column): learners write one thing they now understand about the maize price data that they did not understand before this lesson — specifically referencing grouped data or modal class. On the SECOND note (orange / right column): learners write one NEW question that has emerged from today's lesson — for example: 'If we change the class width, does the modal class always change?' or 'Can two different class widths give the same modal class?' or 'Is the modal class useful for the family making a budget?' Notes are posted on the class DQB under columns labelled 'We Now Understand' and 'We Still Wonder.' The teacher reads 3–4 new questions aloud, annotates which ones will be addressed in Lessons 7–8 (histograms and frequency polygons), and circles any that the class can answer right now using today's evidence.	Say: 'Our Driving Question asks how statistics can tell an honest and useful story. Today we learned that the story can change depending on how we group the data. That is powerful — and a little frightening. Write on your green note: what do you now understand about the maize price story? Write on your orange note: what new question did today's lesson open up for you?' Allow 4 minutes of silent writing. Collect notes and read 4 aloud (anonymously). For each new question, ask: 'Can we answer this right now with what we know, or do we need more evidence?' Mark unanswered questions with a star. Say: 'These starred questions are exactly what Lessons 7 and 8 are designed to answer — the histogram will make the distribution visible in a way the table cannot.'	Metacognitive Charting (NGSS SEP 8 — Obtaining, Evaluating, and Communicating Information): The two-column DQB separates declarative knowledge gains from open inquiry threads, training learners to distinguish what they know from what they still need to investigate. This is a core scientific habit of mind that mirrors how Kenyan data analysts at KNBS (Kenya National Bureau of Statistics) work.	Observable indicator: Green notes contain specific, lesson-linked content (e.g., 'I now understand that the modal class for maize prices is Ksh 50–59, which means most days the family paid between Ksh 50 and Ksh 59'). Vague notes ('I learned about grouping') indicate surface processing — the teacher privately follows up with those learners at the start of Lesson 7. Orange notes that are genuine mathematical questions (not personal comments) indicate productive curiosity.
Phase 5 — Model Building 📹 **VIDEO:** Subtracting with Grouped Pictures Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Grouped Data to Find the Mean (Flexbook) Source: CK-12 🔍 Search ARES for similar	Learners open their cumulative 'Data Story Model' — a running annotated diagram they have been building since Lesson 1. In previous lessons the model included: the raw data list, a stem-and-leaf plot or ungrouped frequency table, and annotations of mean/median/mode. Today, learners ADD a new layer: they sketch their grouped frequency table (Ksh 10 intervals) directly onto their model, draw an arrow from the ungrouped mode to the modal class, and write one annotation answering: 'What does	Say: 'Your model is a living document — it grows every lesson as our evidence grows. Today you are adding a new layer: the grouped picture of the data. Draw your grouped frequency table — it does not need to be beautiful, it needs to be accurate. Then draw an arrow from your ungrouped mode to your modal class and annotate what changed — or didn't.' Circulate and look for models that lack the comparison arrow. Prompt: 'Where is your ungrouped mode from Lesson 3? Connect it to today's modal class	Cumulative Model Revision (NGSS SEP 2 — Developing and Using Models): Revising the same model repeatedly across lessons externalises the learner's growing understanding and makes knowledge-building visible. Connecting ungrouped and grouped representations within the same model diagram helps learners see that both are valid but answer different questions — a key statistical literacy insight.	Observable indicator 1: Learners' models contain the grouped frequency table with correct class intervals and frequencies summing to 40. Observable indicator 2: The annotation on the comparison arrow is specific (e.g., 'The ungrouped mode was Ksh 55, which falls inside the modal class Ksh 50–59 — the grouping confirms the same region is most common'). Observable indicator 3: The updated CBA references both the family AND the neighbour and uses the term 'modal class' or 'grouped data' — indicating

readings	grouping reveal about the maize price distribution that the raw list did not?' Learners then add a second mini-model for Kisumu rainfall — a grouped frequency table with the modal class circled and annotated. At the bottom of their model page, learners write a 'Current Best Answer' (CBA) to the driving question in 2–3 sentences, updating the CBA they wrote in Lesson 5. The teacher circulates and stamps or initials models that show the new grouped table layer and a revised CBA.	— are they in the same interval?' Cold-call 2 learners to read their updated CBA aloud. Ask the class after each: 'Is this a more complete answer than what we had after Lesson 5? What evidence are we still missing?' Point to the DQB starred questions as the bridge to Lesson 7.		integration of today's new concept into the overall argument.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners meaningfully engage with the class-width debate, or did they treat it as a procedural exercise? Evidence: listen back to the gallery-rotation conversations — were learners connecting different class widths to the family-vs-neighbour dispute, or just noting numerical differences? (2) Were the Kisumu rainfall tables completed with the same fluency as the maize price tables, or did transfer to a new context reveal gaps? If most learners struggled with the Kisumu task, plan a 10-minute paired re-attempt at the start of Lesson 7 before introducing histograms. (3) Review the DQB orange notes: do the new questions show that learners are wondering about the visual representation of grouped data (pointing naturally toward histograms in Lesson 7), or are they asking questions about measures of central tendency that should have been resolved in Lessons 3–5? If the latter, the storyline may need a brief consolidation moment at the start of Lesson 7. (4) Count the number of learners whose Data Story Sentence 3 correctly linked the modal class to the family-neighbour dispute. If fewer than 60% achieved this, the Explain phase sensemaking strategy needs strengthening — consider adding a sentence stem scaffold in Lesson 7: 'The modal class of Ksh ___ to Ksh ___ tells us that the [family / neighbour] is [more / less] correct because ___.'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners sorted 40 Wakulima Market maize price cards into class-interval bins using three different class widths (Ksh 5, Ksh 10, Ksh 20) and constructed grouped frequency distribution tables by hand. They performed a gallery rotation comparing all three tables and independently grouped 24 months of Kisumu rainfall data into a second grouped frequency table.
What did I learn?	Learners learned that: (a) grouping raw data into class intervals makes the distribution pattern visible but loses individual values; (b) the modal class is the class interval with the highest frequency — it is a range, not a single number; (c) the choice of class width changes the appearance of the data and can make the same dataset look either 'stable' or 'variable', directly affecting which side of the family-neighbour dispute is supported.
How does this explain the phenomenon?	Learners explained, in written Data Story sentences and updated cumulative models, how the modal class of Ksh 50–59 for the maize price data relates to the ungrouped mode found in Lesson 3, and what this reveals about the typical price a Nairobi family pays at Wakulima Market. They connected the Kisumu rainfall modal class to practical farming decisions, and updated their Current Best Answer to the driving question to incorporate grouped data evidence.

LESSON 7 (80 minutes): Measures of Central Tendency for Grouped Data — Cracking the Maize Price Mystery at Scale	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of this lesson, learners will be able to determine the mean, mode and median of grouped data drawn from the Wakulima Market maize flour price dataset, and use these measures to evaluate competing claims about price stability.
Knowledge	Learners recall that ungrouped mean, mode and median were computed in Lesson 4. They now extend this to grouped frequency distribution tables, learning: (i) the assumed mean (working mean) method for grouped mean; (ii) the modal class and class midpoint as the estimated mode; (iii) the median class located via cumulative frequency, and the interpolation formula Median = $L + [(n/2 - F)/f] \times h$.
Skills	Learners construct a grouped frequency distribution table for the 40 maize prices, compute estimated mean using the assumed-mean/deviation method, identify the modal class, apply the median interpolation formula, and compare all three measures to draw a conclusion about the Wakulima Market price dispute.
Attitudes	Learners appreciate that honest data analysis requires choosing the right measure for the right question; they demonstrate integrity by not cherry-picking a statistic that favours a pre-existing opinion.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	In Lessons 1–6 learners collected the 40 maize prices, organised them into a frequency distribution table, drew a histogram and frequency polygon, and computed mean, mode and median for ungrouped data. Those calculations gave slightly different 'answers' and left a productive tension: the family and the neighbour could both be right depending on which measure you use. Now, with the data fully grouped into class intervals, learners apply the three measures of central tendency to grouped data — the most realistic form in which statistics appear in real Kenyan contexts (KNBS reports, KEBS price bulletins, hospital records). The lesson resolves whether the 'typical' maize price at Wakulima is genuinely stable or variable, moving the class closer to a full answer to the driving question.
Safety Notes	No practical hazards. Ensure calculators are charged/available. If learners share physical worksheets, handle paper tidily to avoid litter.

B. LESSON OVERVIEW

Lesson 7 is the penultimate evidence-gathering lesson in the Statistics I sequence. Learners receive the fully grouped frequency distribution table of 40 Wakulima Market maize flour prices (in Ksh, 8 class intervals of width 5) that was constructed in an earlier lesson. Working in pairs and then as a whole class, they: (1) predict which class holds the 'middle' price; (2) compute the estimated mean using the assumed-mean/deviation method; (3) identify the modal class; (4) apply the median interpolation formula using cumulative frequencies; (5) compare all three measures and revisit both the family's and the neighbour's claims. The lesson uses the NGSS Science and Engineering Practice of Analysing and Interpreting Data (SEP 4) and Constructing Explanations (SEP 6). A Driving Question Board update anchors new vocabulary and lingering questions, and learners revise their cumulative class model of 'what a single number can — and cannot — tell us about a messy real-world dataset.'

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
1 — Predict	Learners receive a printed card	Display the grouped frequency	Prediction Anchoring — by	Teacher scans sticky notes during

📹 **VIDEO:** [Subtracting with Grouped Pictures](#) Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Summary Statistics, Summarizing Univariate Distributions (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	showing the grouped frequency distribution table of the 40 Wakulima Market maize flour prices (classes: 55–59, 60–64, 65–69, 70–74, 75–79, 80–84, 85–89, 90–94 Ksh; frequencies: 2, 4, 7, 11, 8, 5, 2, 1). Before any calculation, each learner silently writes on a sticky note: (a) which class interval they think contains the 'middle' price, (b) whether they think the mean will be higher or lower than the median for this data, and (c) one sentence restating either the family's or the neighbour's position from the anchoring phenomenon. Pairs briefly share predictions (2 minutes). Sticky notes are posted on a 'Before We Calculate' section of the board.	table on the board alongside the original anchoring phenomenon statement: 'Who is right — the family that says prices vary wildly, or the neighbour who says prices are stable?' Say: 'We have organised the 40 prices into groups. Before we do any arithmetic, I want your gut feeling. Look at the table — where do most prices cluster? Write your prediction silently.' Allow 10 seconds of silence (WAIT TIME 1). After writing: 'Turn to your partner — 30 seconds, share your prediction.' Cold-call 3 pairs (select one high-confidence, one uncertain, one who predicted differently). Record the three predictions verbatim on the board under 'Our Predictions.' Say: 'By the end of this lesson, you will be able to tell me whether your prediction was right — and why it matters for the family at Wakulima.'	committing to a prediction before calculating, learners activate prior knowledge of ungrouped measures (Lesson 4) and create cognitive investment. When the calculated answer either confirms or surprises them, the contrast deepens conceptual understanding of why grouping changes (slightly) the results.	the 2-minute pair share. Observable indicator: learner correctly identifies the class 70–74 (frequency 11) as the likely median/modal region. Misconception to watch: learners who confuse 'most frequent class' with 'class containing the mean' — note names for targeted questioning in Phase 3.
2 — Observe 📹 **VIDEO:** [Subtracting with Grouped Pictures](#) Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Summary Statistics, Summarizing Univariate Distributions (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of four. Each group receives Worksheet 7A which contains: (i) the grouped frequency table with a blank cumulative frequency column to complete; (ii) a partially filled assumed-mean deviation table (columns: Class, Midpoint x, Frequency f, Deviation d = x − A, fd — where assumed mean A = 72, the midpoint of the 70–74 class); (iii) a cumulative frequency column to complete for the median calculation. Groups complete the cumulative frequency column and the deviation table (10 minutes). Groups then pass their worksheet one seat to the right for a 2-minute peer check using the answer strip provided by the teacher. Discrepancies are circled and	Distribute Worksheet 7A. Say: 'In Step 1, add a cumulative frequency column — running totals of frequency from the top. In Step 2, fill in the midpoints: the midpoint of 55–59 is 57, of 60–64 is 62, and so on. In Step 3, calculate d = midpoint minus 72 for each row, then multiply by frequency to get fd.' Circulate for 4 minutes. Stop the class: 'Raise your hand if the midpoint of 75–79 is 77.' (Correct answer: 77.) Cold-call one group who did NOT raise their hand — ask them to explain their midpoint and correct collaboratively. Continue circulating. At 8 minutes say: 'Pass your worksheet to the right — check your neighbour's cumulative frequency column using the answer strip. Circle any	Structured Data Completion with Peer Verification (SEP 4 — Analysing and Interpreting Data) — filling in the cumulative frequency and deviation columns is not mechanical copying; it forces learners to see the internal logic of grouped data representation. Peer checking builds metacognitive awareness: learners must articulate why a number is wrong, not just that it is wrong.	Observable indicators: (a) all midpoints correctly calculated as class lower boundary + 2 (e.g., 57, 62, 67, 72, 77, 82, 87, 92); (b) cumulative frequency reaches 40 at the last class; (c) Σfd column has both positive and negative values (deviations below A = 72 are negative). Red flag: any group whose cumulative frequency column does not reach 40 — teacher intervenes immediately before Phase 3.

	brought to whole-class discussion.	discrepancy in red.' After 2 minutes: 'Which group found a discrepancy? What was it?' Cold-call 2 groups. Address the most common error (likely: forgetting to add cumulatively or using class boundaries instead of midpoints).		
3 — Explain 📹 **VIDEO:** [Subtracting with Grouped Pictures] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Summary Statistics, Summarizing Univariate Distributions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Whole-class guided calculation in three sub-phases. Sub-phase A (Mean, 12 min): Using the completed deviation table on the board, the class derives: Estimated Mean = A + (Σfd / n) = 72 + (Σfd / 40). Each group calls out their Σfd; teacher tallies on board to reach consensus (Σfd should be approximately +28, giving mean ≈ 72.7 Ksh). Learners write the full working in their exercise books. Sub-phase B (Modal Class, 5 min): Learners identify the class with the highest frequency (70–74, f = 11) as the modal class and state the modal class midpoint (72) as the estimated mode. Sub-phase C (Median, 13 min): Using the interpolation formula Median = L + [(n/2 − F) / f] × h, learners identify: n/2 = 20; cumulative frequency just below 20 falls in the 65–69 class (cumulative = 13); so the median class is 70–74 (cumulative reaches 24); L = 70 (lower class boundary), F = 13, f = 11, h = 5. Median = 70 + [(20 − 13)/11] × 5 = 70 + 3.18 ≈ 73.2 Ksh. Groups check their own working. Teacher writes all three values on the board: Mean ≈ 72.7 Ksh | Modal class: 70–74 Ksh | Median ≈ 73.2 Ksh. Class discussion: 'Do these three measures agree? What do they tell the family and the neighbour?'	Sub-phase A: Write on board: 'Estimated Mean = A + (Σfd/n).' Say: 'Each group, give me your Σfd.' Record all four group values — if they differ, ask: 'Group 3, walk me through your fd for the 80–84 class.' (WAIT TIME 15 seconds.) Correct any arithmetic error collaboratively. Compute mean together. Sub-phase B: 'Without any formula — just look at the frequency column. Which class has the most prices recorded?' (WAIT TIME 10 seconds.) Cold-call 2 learners. Confirm modal class 70–74. Say: 'The estimated mode is the midpoint of this class — 72 Ksh. Is this close to the mean?' Sub-phase C: Write the median formula on board in full. Say: 'We need n/2. What is half of 40?' (Cold-call 1 learner.) 'Now find the cumulative frequency just BELOW 20 — which class does that land in?' (WAIT TIME 15 seconds, cold-call 1 learner.) Work through L, F, f, h step by step. After the answer: 'Our median is 73.2 Ksh. Our mean is 72.7 Ksh. Our modal class midpoint is 72 Ksh. Turn to your partner — 1 minute — what does this tell the family and the neighbour?' Cold-call 3 pairs to share. Summarise: 'All three measures point to a typical price of about 72–73 Ksh. The NEIGHBOUR is broadly right — the centre is stable. But we still need to explain why the FAMILY felt variation. That question goes	Guided Inquiry with Progressive Reveal (SEP 6 — Constructing Explanations) — by calculating mean, then mode, then median in sequence and comparing them, learners construct the explanation themselves rather than receiving it. The convergence of all three measures around 72–73 Ksh is the 'aha' moment: the data's centre IS stable, which means any variation the family perceived must come from spread (range, variance) — a concept deliberately left as a lingering question to motivate future lessons.	Observable indicators: (a) learner correctly substitutes L = 70, F = 13, f = 11, h = 5 into the median formula with no prompting; (b) learner states the median class (not just the median value) correctly; (c) during pair discussion, learner connects the numerical result back to the phenomenon ('the neighbour is right because all three measures give about 72–73 Ksh'). Misconception to watch: learners using class boundaries 69.5–74.5 instead of stated boundaries 70–74 for L — address explicitly: 'In KICD problems at this level, use the stated lower boundary unless told otherwise.'

		on our Driving Question Board.'		
4 — Driving Question Board (DQB) Update 📹 **VIDEO:** Subtracting with Grouped Pictures Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner takes two sticky notes. On the GREEN note they write one thing they can now confidently say about the Wakulima maize price data using today's vocabulary (mean, modal class, median, interpolation). On the ORANGE note they write one question that today's lesson raised but did not answer (e.g., 'Why did the family still feel variation if the centre is stable?' / 'What if the class widths were unequal?' / 'Is 72–73 Ksh a fair price — who decides?'). Green notes go under 'What We Now Know'; orange notes go under 'What We Still Wonder.' Teacher reads three orange notes aloud and circles the one most relevant to the next lesson (on histograms with unequal class widths and frequency density). Teacher updates the class's cumulative 'Answer Thread' on the DQB: adds the arrow from 'Lesson 7' pointing to 'The centre of the maize price distribution is approximately 72–73 Ksh — the neighbour's claim is statistically supported.'	Distribute sticky notes. Say: 'You have 3 minutes — silent writing. Green: one confident claim with a number from today. Orange: one honest question today did not answer.' (WAIT TIME: full 3 minutes of silence.) Collect boards. Read three orange notes aloud without revealing authors. Ask: 'Which of these questions do you think is MOST important for us to answer next?' (Quick show of hands.) Circle the winning question on the DQB. Say: 'Next lesson we will look at histograms where the classes are different widths — like a maize price chart from a KNBS report — and that will help us answer this orange question.' Point to the cumulative Answer Thread and add today's arrow visibly in front of learners.	Driving Question Board Protocol — the two-colour sticky note routine externalises both consolidation and curiosity simultaneously. By physically placing notes on a shared board, learners see that the class's collective understanding is growing lesson-by-lesson, building epistemic momentum and a sense of shared scientific community.	Observable indicator: green notes contain at least one correct numerical value (e.g., 'The mean maize price is about 72.7 Ksh') AND use correct vocabulary (mean/median/modal class). Red flag: green notes that say only 'I learned the formula' without a number — teacher notes these learners for follow-up in the next lesson's opening review.
5 — Model Building 📹 **VIDEO:** Subtracting with Grouped Pictures Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook)	Learners retrieve their cumulative class model — a large annotated diagram begun in Lesson 1 that has grown each lesson. The model currently shows: raw data list → tally/frequency table → histogram → frequency polygon → ungrouped mean/mode/median. Today learners add a new layer: they annotate the grouped frequency table section of the model with arrows labelled 'Estimated Mean = 72.7 Ksh (assumed-mean method)', 'Modal Class = 70–74 Ksh', and 'Median ≈	Say: 'Open your cumulative model to the grouped frequency table section. You have 7 minutes to add today's three results and write the Verdict Box. Use the exact phrasing on the board for the Verdict Box.' (WAIT TIME: 7 minutes working time.) At 5 minutes: 'You have 2 minutes — swap with your partner and add anything you missed in a different colour.' After swap: cold-call 2 learners to read their Verdict Box aloud. Offer specific, corrective praise: 'Excellent — Amina used	Cumulative Model Revision (SEP 2 — Developing and Using Models) — the model is not a one-lesson artefact but a living document that grows with the evidence sequence. Adding a 'Verdict Box' forces learners to synthesise across all seven lessons and articulate a conclusion in plain language, mirroring how a statistician or a KNBS analyst would present findings to a non-technical audience.	Observable indicator: the Verdict Box correctly distinguishes between 'centre' (what today's lesson addresses) and 'spread' (what a future lesson will address), demonstrating that learners understand the limitation of central tendency measures alone. Red flag: any model that says 'the data shows the prices are always 72–73 Ksh' — this conflates the measure of centre with individual values and must be corrected before Lesson 8.

Source: CK-12 🔍 Search ARES for similar readings	73.2 Ksh (interpolation)'. They also write a 'Verdict Box' in the centre of their model: 'The centre of the Wakulima maize price distribution is approximately 72–73 Ksh. The neighbour's claim of price stability is supported by all three measures of central tendency. The family's experience of variation is not about the centre — it must be about SPREAD.' Learners compare their model with a partner and add one annotation the partner had that they missed.	the word interpolation correctly. Baraka, your Verdict Box says the mean is 72 — can you add the decimal to make it more precise?' Close: 'Your model now tells a story that goes all the way from a chaotic list of 40 prices to a single sentence that settles a real argument between a family and their neighbour. That is what statistics does — it turns noise into knowledge.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. CONCEPTUAL DEPTH: Did learners understand WHY we use assumed mean A = 72 (or any convenient value) — i.e., that it simplifies arithmetic without changing the answer — or did they treat it as a magic rule? If the latter, plan a 5-minute 'What if we chose A = 67?' activity at the start of Lesson 8.
2. FORMULA FLUENCY: How many learners correctly identified L, F, f, and h in the median formula without prompting? If fewer than 70% did so, consider a paired 'formula detective' card sort at the start of Lesson 8 where learners match formula components to their definitions.
3. PHENOMENON CONNECTION: Did learners spontaneously connect the three computed values back to the family vs. neighbour dispute, or did you have to prompt every connection? The goal is for learners to make the link unprompted by Lesson 8.
4. EQUITY CHECK: Were there learners who disengaged during the calculation phases? Statistics can feel alienating when it becomes purely procedural. Note whether the Wakulima context kept motivation high across ability levels — if not, consider sourcing a second real dataset (e.g., Kisumu ferry fares or Kericho tea picker daily wages) for differentiated extension in Lesson 8.
5. MODEL QUALITY: Review 4–5 cumulative models before the next lesson. Check that the Verdict Box correctly separates 'centre' from 'spread.' This distinction is the conceptual bridge to Lesson 8 (histograms with unequal class widths) and ultimately to standard deviation in Grade 11.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed the grouped frequency distribution table of 40 Wakulima Market maize flour prices organised into 8 class intervals, completed the cumulative frequency column, and filled in a deviation table using an assumed mean of 72 Ksh.
What did I learn?	Learners learned how to estimate the mean of grouped data using the assumed-mean/deviation method, identify the modal class and its midpoint as the estimated mode, and apply the median interpolation formula (Median = L + [(n/2 − F)/f] × h) to find the median class and estimated median value.
How does this explain the phenomenon?	Learners explained that all three measures of central tendency — mean (≈72.7 Ksh), modal class midpoint (72 Ksh) and median (≈73.2 Ksh) — converge around 72–73 Ksh, supporting the neighbour's claim that Wakulima maize prices are broadly stable. The family's perceived variation must therefore be about the spread of prices around that centre, not about the centre itself — a question that drives the next lesson.

LESSON 8 (40 minutes): Which Number Tells the Truth? — Final Explanation: Mean, Mode and Median of the Maize Price Data	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners calculate mean, mode and median for the Wakulima Market maize price data, discover how a price spike distorts the mean, compare all three measures, and produce a final evidence-based verdict on the family-vs-neighbour dispute — completing the sub-strand storyline.
Knowledge	– The mean of ungrouped data is calculated as $\Sigma fx \div \Sigma f$; an extreme value (outlier) pulls the mean away from where most values cluster. – The median is the middle value when data is ordered; it is resistant to outliers. – The mode is the most frequently occurring value; it reflects the 'everyday' price experienced by a typical shopper. – The choice of which measure to report is itself a decision that can make data tell an honest or misleading story.
Skills	– Calculate mean, mode and median by hand from a frequency distribution table of ungrouped data. – Evaluate and justify which measure of central tendency best represents a given real-life data set. – Construct a written statistical argument that uses evidence to resolve a real-world dispute.
Attitudes	– Appreciate that honesty in data reporting requires deliberate choice of appropriate statistical measures. – Value the power of mathematics to resolve real community disagreements fairly and transparently.
Key Inquiry Question	How do we use statistics in day-to-day life — and which summary number tells the most honest story about data from our Kenyan community?
Purpose in Storyline	This is the final lesson of the sub-strand. Learners bring together every tool built across Lessons 1–7 — frequency tables, histograms, grouped and ungrouped measures — and apply them to produce a complete, justified verdict on the Wakulima Market Maize Price Mystery, directly answering the driving question.
Safety Notes	No specific hazards. Ensure learners handle calculators responsibly and that written work is their own during individual calculation tasks.

B. LESSON OVERVIEW

This lesson is the culminating explanation lesson for Sub-Strand 3.1: Statistics I. Learners return, for the final time, to the anchoring phenomenon — the Wakulima Market Maize Price Mystery — and use the complete toolkit assembled over the previous seven lessons to compute mean, mode and median from the ungrouped maize price frequency table. The central cognitive challenge of the lesson is the mandazi canteen supporting phenomenon: the teacher presents a small hypothetical data set of daily mandazi sales prices at the school canteen in which one extraordinary price spike (caused by a sugar shortage in Rift Valley) dramatically inflates the mean while the median and mode remain stable. This concrete, relatable scenario creates productive cognitive conflict — learners discover that the mean can deceive, and that choosing which statistic to report is itself an ethical, mathematical decision.\n\nIn the Explain phase learners work in pairs to calculate all three measures for the full 40-price maize data set, compare them, and write a one-paragraph verdict answering: 'Who was right — the family or the neighbour?' The structured verdict writing is supported by a sentence-frame scaffold. Crucially, learners discover that all three people could be simultaneously 'telling the truth' depending on which measure they were unconsciously experiencing — a powerful realisation about the social responsibility embedded in statistical reporting.\n\nThe lesson closes with a DQB completion ceremony: every question card posted across the eight-lesson unit is revisited, answered with evidence, and the board is declared 'closed'. In the Model Building phase learners revise their personal statistical model (first sketched in Lesson 1) into a final annotated summary diagram showing the full data journey — from raw chaos to frequency table to histogram to measures of central tendency — and write one sentence connecting each tool to the driving question. This final model serves as the

primary summative artefact for the sub-strand.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Mean and Standard Deviation of Discrete Random Variables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a slip showing five daily prices of mandazi at the school canteen: KSh 5, 5, 5, 5, and 45 (the KSh 45 spike occurred when the canteen owner had to use expensive imported sugar during a shortage). Without calculating, learners write in their notebooks which single number — mean, median, or mode — they think best represents the 'typical' daily mandazi price, and why. They then share their prediction with a shoulder partner for 90 seconds.	Distribute the mandazi price slips face-down. Say: 'Do not turn over until I say go.' After slips are turned over, give 2 minutes silent individual writing time. Then say: 'Turn to your shoulder partner. You have 90 seconds — tell them which measure you chose and your reason. Go.' After 90 seconds, cold-call 3 learners using name sticks: 'Kamau — which did you choose and why?' 'Akinyi — do you agree with Kamau or did you choose differently?' 'Mutua — is there a right answer here, or could different people be correct?' Allow 10 seconds of WAIT TIME after each question before accepting a response. Record the three choices (mean / median / mode) as a tally on the board — this tally will be revisited after the Observe phase.	Anchoring Prediction with a Supporting Phenomenon — By using the mandazi scenario (smaller, more manageable numbers than the 40-price maize list) learners activate prior knowledge of all three measures in a low-stakes context before applying them to the full phenomenon. Recording the class tally creates a public commitment that generates genuine curiosity about the outcome.	Listen for: Do learners correctly recall how mean, median and mode are each calculated? Can they articulate why an extreme value might affect one measure more than another? Watch for learners who default to 'mean is always best' — flag these learners for targeted questioning during the Explain phase. Target indicator: at least 70% of learners can name a reason related to the spike price in their written prediction.
Observe Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Mean and Standard Deviation of Discrete Random Variables	Learners work in pairs to calculate the mean, median and mode of the five mandazi prices (5, 5, 5, 5, 45) by hand, showing full working. They then record the three results in a comparison table drawn in their notebooks with columns: Measure | Value (KSh) | Does this feel like the 'typical' price? (Yes / No / Partly). After completing the mandazi mini-task, each pair opens their existing frequency distribution table for the Wakulima Market maize prices (built and used in Lessons 3–4 and referenced in Lessons 5–7) and	While pairs work on the mandazi calculation, circulate and use the following diagnostic questions (pose quietly to individual pairs, not whole-class): 'Show me how you are finding the mean — write out Σfx.' 'For the median — how many values do you have? So which position is the middle?' 'What is your mode — and why?' After 5 minutes, pause the class. Cold-call one pair to share their mandazi comparison table on the board. Ask the class: 'Hands up if the mean surprised you.' Count hands aloud: 'I see 14 hands —	Scaffolded Parallel Tasks — Learners first work with the simple 5-value mandazi dataset (where the distortion is obvious) before transferring the same three procedures to the 40-value maize dataset. This parallel structure makes the concept of outlier distortion salient before learners encounter it in the more complex context, reducing cognitive load while preserving the discovery moment.	Correct mandazi results: mean = KSh 13, median = KSh 5, mode = KSh 5. For maize prices: mean ≈ KSh 105.10, median = KSh 103 (20th and 21st values), mode = the most frequent price in the table (to be confirmed from learners' own frequency tables built in Lesson 3). Watch for: (i) learners averaging the five mandazi values without weighting (error: 65÷5=13 is actually correct here since no repeats require weighting, but some may confuse method); (ii) learners unable to locate the 20th/21st values in the maize

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	calculates the mean using Σfx ÷ Σf, identifies the mode (highest frequency value), and lists the ordered values to locate the median.	that is interesting. Why might the mean surprise us here?' Allow 15 seconds WAIT TIME. Then say: 'Now apply exactly the same three calculations to your maize price frequency table. You have 8 minutes. If you need the Σfx value from Lesson 4, it is written on the board.' Write on board: 'Σfx = 4,204 (from Lesson 4); Σf = 40.' Monitor closely — pause after 8 minutes for a mid-task check: 'Hold up your notebooks — show me your mean calculation.' Visually scan for systematic errors in median positioning.		ordered list — prompt with 'your frequency table already has a running total column — use it.' Success indicator: 80% of pairs produce a correct mean for the maize data within 8 minutes.
Explain Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Mean and Standard Deviation of Discrete Random Variables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner independently writes a 'Statistical Verdict' — a structured paragraph of 5–7 sentences answering the driving question: 'Who was right — the Nairobi family or the neighbour?' using the sentence-frame scaffold provided. They must cite all three measures (mean, median, mode) by name and value, explain why the mean is higher than the median (linking to the Rift Valley supply shortage spike visible in the data), and state which measure they recommend the family use for weekly budgeting — with justification. After writing, pairs compare verdicts and identify one point of agreement and one difference in their reasoning. Two pairs are cold-called to read their verdict aloud to the class.	Display the sentence-frame scaffold on the board: 'The mean price of maize flour at Wakulima Market is KSh ___, which suggests ___. However, the median is KSh ___, meaning ___. The mode is KSh ___, which shows ___. The mean is higher than the median because ___. For the family's weekly budgeting, I recommend using the ___ because ___. Therefore, [family / neighbour / both] was right because ___.' Give learners 7 minutes of silent individual writing. After writing, say: 'Compare with your partner — circle one sentence in their verdict that you think is especially strong.' After 2 minutes, cold-call Pair A: 'Read your verdict aloud.' After they read, ask the class: 'Do you agree that [measure they recommended] is best for budgeting? Raise your hand if you agree.' Then cold-call Pair B for a different perspective. Follow up with the whole class: 'Can BOTH the family AND the neighbour be telling the truth at the same time?' Allow 15 seconds WAIT TIME. Accept 3 responses.	Structured Academic Controversy via Verdict Writing — The sentence-frame scaffold ensures all learners engage with all three measures and with the causal explanation (Rift Valley shortage → price spike → mean inflation) rather than selecting only the measure that supports their prior prediction. Comparing verdicts in pairs surfaces the legitimate disagreement between measures, reinforcing that statistical interpretation requires judgement, not just calculation.	Collect and scan 6–8 randomly selected written verdicts during pair-comparison time. Look for: (i) correct numerical values cited for all three measures; (ii) a causal explanation linking the spike price to mean distortion; (iii) a justified recommendation (median or mode is more appropriate than mean for budgeting when outliers are present). Red-flag responses: any verdict that concludes only one party is right without acknowledging the legitimacy of the other's experience — use these as teaching moments in the DQB phase. Target: 75% of verdicts meet all three criteria.

		Conclude by saying: 'Statistics does not always give one truth — it gives us tools to describe different aspects of reality honestly. That is what makes it powerful and also what makes it dangerous if misused.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Mean and Standard Deviation of Discrete Random Variables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher facilitates a whole-class DQB Completion Ceremony. The full driving question board — which has accumulated question cards, partial-answer sticky notes, and revision arrows across all eight lessons — is displayed at the front. Learners review every card and, for each one, vote with thumbs: up (fully answered), sideways (partially answered), down (still open). For the three KICD Key Inquiry Questions (What is statistics? / How do we represent data? / How do we use statistics in day-to-day life?) each learner writes a final one-sentence answer in their notebook. The class then collectively agrees on a 'class consensus answer' for the driving question, which the teacher writes on a fresh card and places at the centre of the board.	Stand beside the DQB. Point to the driving question card at the top: 'Eight lessons ago, I put this question up and you said you had no idea how to answer it. Let us see how far we have come.' Work through the board systematically — hold up each question card and ask for a thumbs vote. For each 'thumbs sideways' card, ask: 'What do we now know about this, and what is still missing?' For the three KICD Key Inquiry Questions, say: 'Everyone — in your notebook, write your own final answer to this question in one sentence. You have 90 seconds.' After 90 seconds, cold-call 3 different learners (choosing learners who have not been cold-called in this lesson yet): 'Otieno — what is your answer to: How do we use statistics in day-to-day life?' After 3 responses, synthesise: 'So we can say statistics helps us [summarise / represent / interpret] data so that we and our communities can make better decisions — even when the raw numbers look chaotic.' Write the class consensus answer to the driving question aloud as you write it on the board: 'We can use statistics to tell an honest story about data from our community by organising it into frequency tables, displaying it in histograms, and choosing the right measure of central tendency — mean, median, or mode — depending on what	DQB Closure Ritual — Systematically revisiting every question posted since Lesson 1 gives learners a tangible sense of intellectual progress and models the scientific practice of tracking how explanations evolve with evidence. The thumbs-vote creates low-stakes public accountability, and writing a final individual answer before hearing the class consensus ensures every learner articulates their own understanding rather than copying a peer.	Teacher scans notebook entries for the three KICD Key Inquiry Question answers during the 90-second writing pause. Look for: specificity (does the learner mention specific tools — frequency table, histogram, mean/median/mode — or remain vague?), accuracy (are any misconceptions still present?), and connection to the phenomenon (does the learner reference maize prices, Wakulima Market, or the family/neighbour dispute?). Any learner still giving a vague answer ('statistics helps us understand numbers') should be flagged for a brief teacher check-in during the Model Building phase.

		question we are trying to answer and who might be harmed by a misleading number.'		
Model Building Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Mean and Standard Deviation of Discrete Random Variables (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner revises their personal statistical model — the annotated diagram first sketched in Lesson 1 as a chaotic list of 40 numbers with a question mark. Using their full set of lesson notes, they now complete the model by drawing a data-journey flowchart with five labelled stations: (1) Raw Data — 40 maize prices from Wakulima Market; (2) Frequency Distribution Table — organised, tallied, frequencies summed; (3) Histogram — shape and distribution visible, right skew explained; (4) Grouped Table and Modal Class — pattern at scale; (5) Measures of Central Tendency — Mean KSh 105.10, Median KSh 103, Mode [class value]. At each station they write one connecting sentence linking that tool back to the driving question. They annotate the mean station with a red asterisk and the note: 'Caution — sensitive to outliers (Rift Valley shortage spike).' Finally, at the bottom of the model, they write their Statistical Verdict in one sentence.	Distribute the Model Building template (blank A4 sheet with five numbered boxes in a flowchart arrangement and the prompt: 'At each station, answer: What does this tool show us about the Maize Price Mystery?'). Say: 'This is your final model for Sub-Strand 3.1. Everything we have done over eight lessons lives inside this one diagram. You have 8 minutes. Work independently and in silence.' Circulate and use targeted prompts for learners who are stuck: 'What did the histogram tell us that the raw list could not?' 'Why did we need the median as well as the mean?' 'What would you tell the family to use when budgeting — and why?' In the final 2 minutes, ask learners to swap models with a partner and check: 'Can you follow their data journey from raw data to verdict? If yes, put a tick. If something is missing, draw an arrow and write a question.' Collect all models as the summative artefact for Sub-Strand 3.1.	Cumulative Model Revision — Comparing the Lesson 1 model (a chaotic list with a question mark) to the Lesson 8 final flowchart makes the learner's own intellectual growth concrete and visible. The peer-check with arrows and questions embeds the scientific practice of peer review and supports self-regulation. Annotating the mean station with a caution flag reinforces the ethical dimension of statistical reporting, directly fulfilling the KICD value of integrity.	Collect all final models. Use a 3-point holistic rubric: (3) All five stations present, each with a correct connecting sentence and correct numerical values; caution note on mean; clear Statistical Verdict. (2) Four of five stations correct; verdict present but may lack full justification. (1) Fewer than four stations or missing verdict or persistent numerical error. Target: 80% of learners at level 2 or above. Flag all level-1 models for follow-up consolidation activity. Also check that the Lesson 1 model (if retained in notebooks) shows visible revision — this evidences metacognitive growth across the sub-strand.

D. TEACHER REFLECTION

1. Did learners successfully connect the mandazi canteen spike to the maize price outlier — and articulate in their own words WHY the mean is more sensitive to extreme values than the median or mode? If not, what alternative concrete example could I use next time to make the outlier-distortion concept more visceral?

2. When learners wrote their Statistical Verdicts, did the majority recommend median or mode for budgeting, or did a significant group still prefer the mean? What does this reveal about whether the concept of 'choosing the right measure' has genuinely transferred, versus learners applying a rote rule?

3. During the DQB Completion Ceremony, were there any question cards that remained genuinely unanswered — questions the class posted that we never returned to? How should I address these at the start of Sub-Strand 3.2 (Probability)?

4. Looking at the final model artefacts: did learners draw a coherent data-journey from raw data to verdict, or did they produce disconnected summaries of each lesson? What does this tell me about whether the storyline structure across all eight lessons succeeded in building cumulative understanding?

5. Did I allocate enough time for the DQB Closure Ritual without rushing the Model Building phase? In future, would splitting these two phases across two lessons — or using a 70-minute double period for Lesson 8 — produce better quality final models and a less rushed verdict-writing experience?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that when a single extreme price (KSh 45 mandazi spike; analogous to the Rift Valley-shortage maize price spike) is included in a data set, the mean is pulled noticeably higher than both the median and the mode — even though the extreme value occurred only once. They also observed, by completing the DQB Closure Ceremony, that every question posted across eight lessons now has an evidence-based answer.
What did I learn?	The mean of ungrouped data ($\Sigma fx \div \Sigma f$) is sensitive to outliers and can misrepresent the 'typical' value experienced by most people. The median and mode are more resistant to extreme values. The choice of which measure to report is a deliberate decision with real consequences for how data tells its story — and for whether that story is honest. For the Wakulima Market maize price data: mean ≈ KSh 105.10, median ≈ KSh 103, and the mode reflects the most commonly charged price; the family budgeting for weekly shopping should use the median or mode for a realistic typical-day estimate.
How does this explain the phenomenon?	Both the Nairobi family and their neighbour were telling the truth as they experienced it. The neighbour, shopping on typical days, encountered prices clustered near the mode and median — so prices felt 'always about the same.' The family, whose data included days with a Rift Valley supply-shortage spike, experienced genuine price variation — and their mean was inflated by those rare high-price days. Statistics resolves the mystery not by declaring one person wrong, but by showing that different measures capture different aspects of reality — and that choosing the right measure depends on the question you are trying to answer and the decisions that rest on it.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.